MA411 Notes 4

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9 Conditional expectation

We saw informally conditional expectation $\mathbb{E}(X|A)$ where X is a random variable and A is an event before.

Definition 1 (conditional expectation: on an event)**.** *Let $\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and $X : \Omega \to \mathbb{R}$ be a random variable such that $\int_\Omega |X| \, \mathrm{d}\mathbb{P} < \infty$. If $A \in \mathcal{F}$ has $\mathbb{P}(A) > 0$, we define the* conditional expectation $\mathbb{E}(X|A) = \mathbb{P}(A)^{-1} \int_A X \, \mathrm{d}\mathbb{P}$*, if it exists.*

This is only really different from the previous definition in that we now formally know how to evaluate the expectation of a general random variable. Recall that the Lebesgue integral generalises the sum definition we gave before in the special case that Ω is a finite set.

Expectation and conditional expectation don't behave well if we drop the condition $\int_\Omega |X| \, \mathrm{d}\mathbb{P} < \infty$ (sometimes written 'X is in L^1'), so we will routinely insist on this.

Example 2. *Consider tossing two fair coins in sequence. Let X count the number of Heads obtained in total, and let Y be the indicator of the event that the first coin lands Heads.*

We can make sense of the conditional expectations $\mathbb{E}(X|Y = 0) = 2\big(0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4}\big) = \frac{1}{2}$ and $\mathbb{E}(X|Y = 1) = 2\big(1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{4}\big) = \frac{3}{2}$.

But it would be 'nicer' to write $\mathbb{E}(X|Y) = Y + \frac{1}{2}$, by which we intend to capture both conditional expectations in one formula. If we know the value of Y, then the conditional expectation of X is $Y + \frac{1}{2}$.

We should immediately notice: $\mathbb{E}(X|Y)$ is *not a number*. It is a random variable: a function from Ω to $\mathbb{R}$.

This immediately means we should put in a sanity condition: if $\mathbb{E}(X|Y)$ is supposed to be our best guess at the value of X given the information Y, then we should certainly give the *same* guess for $\omega, \omega' \in \Omega$ if $Y(\omega) = Y(\omega')$. In other words, $\mathbb{E}(X|Y)$ should be $\sigma(Y)$-measurable. In this example, that happened 'automatically': we decided the right function to write is $Y + \frac{1}{2}$, and that function is $\sigma(Y)$-measurable.

Example 3. *Consider making two independent uniform samples from $[0, 1)$ in sequence. Let X be the sum of the samples, and let Y be the first sample.*

Intuitively, we would like to say: for any given $y \in [0,1)$, if the first sample is $Y = y$, our best guess for the sum of the two samples is $y + \frac{1}{2}$; it seems to be reasonable to call that the conditional expectation.

So we would like to write (again) $\mathbb{E}(X|Y) = Y + \frac{1}{2}$. As in the previous example, this function is $\sigma(Y)$-measurable, which (for the same reason) is an obvious 'sanity check'.

Unfortunately, this time we cannot make sense of this conditional expectation in terms of the event $Y = y$: that event has probability 0.

However, we don't really ever want to know what happens for the event $Y = y$, because that event has probability 0: it doesn't actually happen! What we do need is something we can integrate over events of positive probability and expect to get the right answer, because those are events that actually happen.

This motivates the following formal definition.

Definition 4 (conditional expectation: on a random variable or σ-algebra)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let X be a real-valued random variable on Ω, and Y be an $(S, \mathcal{S})$-valued random variable on Ω. Suppose that $\int_\Omega |X| \, \mathrm{d}\mathbb{P} < \infty$. Let $\mathcal{G} = \sigma(Y)$.*

Suppose that Z is a real-valued random variable satisfying:

(i) $\int_\Omega |Z| \, \mathrm{d}\mathbb{P} < \infty$,

(ii) Z is $\mathcal{G}$-measurable, and

(iii) For all $A \in \mathcal{G}$ we have $\mathbb{E}(Z\mathbf{1}_A) = \mathbb{E}(X\mathbf{1}_A)$.

Then we say Z is a version of $\mathbb{E}(X|Y)$*; we write '$Z = \mathbb{E}(X|Y)$ almost surely'. We may also (equivalently) write $\mathbb{E}(X|\mathcal{G})$.*

Observe that Y doesn't actually play a rôle in the definition except to define $\mathcal{G} = \sigma(Y)$. This justifies writing $\mathbb{E}(X|\mathcal{G})$. There is no loss of generality in assuming that $\mathcal{G}$ is generated by a random variable: for any $\mathcal{G} \subseteq \mathcal{F}$ a σ-algebra, there is a (not necessarily real-valued) random variable Y with $\sigma(Y) = \mathcal{G}$. For example one can take the identity function from Ω to $(S, \mathcal{S})$ with $S = \Omega$ and $\mathcal{S} = \mathcal{G}$.

While in simple examples (such as above) we usually want to think of conditioning on a random variable (we are told the value of Y: what is now our best guess for X?), in more complicated settings it's notationally easier to condition on a σ-algebra (remember this too is conveying information!), and so this is what we'll usually do.

To understand the conditions, (i) is our 'good behaviour' requirement: we don't want to deal with infinities. Condition (ii) is the 'sanity check': intuitively, it's telling us we aren't allowed to make different guesses for the value of X at $\omega, \omega' \in \Omega$ if the value of Y at those two points is the same. Finally, (iii) is 'we can integrate and get the right answer': by definition of the expectation, integrating is exactly what we're doing.

One needs to think a bit about why we allow integrating over $\mathcal{G}$-measurable events. Why not only over Ω, or over all of $\mathcal{F}$?

Integrating only over Ω is not restrictive enough: it would allow us (for instance) to just take $Z = \mathbb{E}(X)$ to be a constant function: but we 'should' be taking into account the value of Y!

On the other hand, integrating over any $\mathcal{F}$-measurable event is unrealistically restrictive: in the two examples above, it would rule out any function being a 'conditional expectation'. In the two-coins Example 2, it would say we should be able to integrate over $A = \{(H, H)\}$ to get $\frac{1}{2}$ (on this set we have $X = 2$, and the probability of the set is $\frac{1}{4}$). But it would also say we should get $\frac{1}{4}$ integrating over $A = \{(H, T)\}$. And this cannot be: our 'sanity check' (ii) says we must make the same guess for both (H, H) and (H, T), because Y is the same on both points. In general, integrating over events not in $\mathcal{G}$ is expecting us to take account of information we don't yet have.

We can think of the condition of integrating over $\mathcal{G}$-measurable events as requiring we take account of all the information we have, but (as the sanity check (ii) requires) not asking us to foretell information we don't yet have.

Finally, we need to explain the 'a version of' bit. In the examples above, there was 'one obvious choice' of the conditional expectation. But actually, in Example 3 there were other choices. For example

$$Z' = \begin{cases} Y + \frac{1}{2} & \text{if } Y \neq 0.3 \\ 17 & \text{if } Y = 0.3 \end{cases} .$$

is a version of the conditional expectation. Changing the number on a ($\mathcal{G}$-measurable) set of measure zero doesn't affect the integration condition (iii). In more complicated examples, sometimes there isn't 'an obvious choice' for the conditional expectation.

Exercise 1. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and X a random variable with $\mathbb{E}(|X|) < \infty$. Suppose $\mathcal{G}$ is a σ-algebra on Ω with finitely many sets, and suppose $\mathbb{P}(G) > 0$ for all $G \in \mathcal{G}$ with $G \neq \emptyset$.*

Prove that there is exactly one function Y such that $Y = \mathbb{E}(X|\mathcal{G})$. (Hint: consider the minimal non-empty sets in $\mathcal{G}$: how should Y be defined on these?)

Solution to Exercise 1:

We first define a function that works: given $\omega \in \Omega$, consider the collection of all sets in $\mathcal{G}$ which contain ω. This is a finite collection, so the intersection of all these sets is a set F_ω in $\mathcal{G}$, and the set is non-empty since it contains ω. By assumption, $\mathbb{P}(F_\omega) > 0$, and so $\mathbb{E}(X|F_\omega)$ is defined. Let $Y(\omega) = \mathbb{E}(X|F_\omega)$. Equivalently, $Y = \sum_A \mathbf{1}_A \mathbb{E}(X|A)$ where the sum runs over all minimal non-empty sets in $\mathcal{G}$.

We claim this function is a version of $\mathbb{E}(X|\mathcal{G})$. For (i), observe that if for some minimal non-empty $A \in \mathcal{G}$ we have $\mathbb{E}(X|A)$ equal to ∞ or $-\infty$, then $\mathbb{E}(|X|) \geq \mathbb{E}(|X|\mathbf{1}_A) = \infty$. It follows these numbers are all real numbers: and now $\mathbb{E}(|Y|) = \sum_A \mathbb{P}(A)|\mathbb{E}(X|A)|$ is a finite sum of real numbers so is finite. The $\mathcal{G}$-measurability (ii) is true by construction: we wrote Y as a simple function with $\mathcal{G}$-measurable indicators. Finally, given $B \in \mathcal{G}$, we have

$$\mathbb{E}(Y\mathbf{1}_B) = \mathbb{E}\big(\mathbf{1}_B \sum_A \mathbf{1}_A \mathbb{E}(X|A)\big) = \mathbb{E}\Big(\sum_{A \subseteq B} \mathbf{1}_A \cdot \tfrac{\mathbb{E}(X\mathbf{1}A)}{\mathbb{P}(A)}\Big) = \sum_{A \subseteq B} \mathbb{P}(A) \cdot \tfrac{\mathbb{E}(X\mathbf{1}A)}{\mathbb{P}(A)} = \sum_{A \subseteq B} \mathbb{E}(X\mathbf{1}_A) = \mathbb{E}(X\mathbf{1}B)$$

as required for (iii). Here all the sums over A run over A a non-empty minimal set in $\mathcal{G}$ (where indicated, contained in B); in particular, that means the sums are over disjoint sets.

Now suppose Y' is also a version of $\mathbb{E}(X|\mathcal{G})$. The set $\{\omega : Y(\omega) > Y'(\omega)\}$ is a measurable set (because $Y - Y'$ is measurable), so let it be $B \in \mathcal{G}$. If B is non-empty, take a minimal non-empty subset A of B. By definition $Y - Y'$ is constant on A, for if not it would contradict $\mathcal{G}$-measurability, and it must take some strictly positive value c. But then $\mathbb{E}(Y\mathbf{1}_A) = c\mathbb{P}(A) + \mathbb{E}(Y'\mathbf{1}_A) > \mathbb{E}(Y'\mathbf{1}_A)$, which contradicts the fact that both sides must equal $\mathbb{E}(X\mathbf{1}_A)$ by condition (iii). Thus $B = \emptyset$; by symmetry the set of ω such that $Y(\omega) < Y'(\omega)$ is empty, and we have $Y = Y'$ everywhere.

In general there can be many versions of the conditional expectation.

Exercise 2. *Suppose Y and Y' are two versions of the conditional expectation $\mathbb{E}(X|\mathcal{G})$. Prove that $Y = Y'$ almost surely.*

Solution to Exercise 2:

Consider the $\mathcal{G}$-measurable function $Z = \max(Y - Y', 0)$. Consider for $n \in \mathbb{N}$ the set $A_n = \{\omega : Z(\omega) \geq \frac{1}{n}\}$. By definition we have $A_n \in \mathcal{G}$, and hence applying condition (iii) twice,

$$\mathbb{E}(X\mathbf{1}A_n) = \mathbb{E}(Y\mathbf{1}_{A_n}) \geq \mathbb{E}\big(Y'\mathbf{1}_{A_n} + \tfrac{1}{n}\mathbf{1}_{A_n}\big) = \mathbb{E}(Y'\mathbf{1}_{A_n}) + \tfrac{1}{n}\mathbb{P}(A_n) = \mathbb{E}(X\mathbf{1}_{A_n}) + \tfrac{1}{n}\mathbb{P}(A_n)\,.$$

It follows $\mathbb{P}(A_n) = 0$, and taking the countable union, $\mathbb{P}\big(\bigcup_{n\in\mathbb{N}} A_n\big) = 0$. Now this is the set of ω such that $Y(\omega) > Y'(\omega)$; by symmetry also the set of ω such that $Y(\omega) < Y'(\omega)$; has measure zero, proving $Y = Y'$ almost surely.

This exercise justifies writing '$Y = \mathbb{E}(X|\mathcal{G})$ almost surely'.

Another way of thinking of the conditional expectation $\mathbb{E}(X|\mathcal{G})$ is that it is the 'closest' random variable Y to X which is $\mathcal{G}$-measurable. Here 'closest' is defined by the 'L_2-distance' $\mathbb{E}\big((X - Y)^2\big)$. Of course, this only makes sense if the distance is defined; for example, if $\mathbb{E}\big(X^2\big) < \infty$.

It's worth noting that these expectation conditions form a hierarchy: if for some $n \geq 2$ we have $\mathbb{E}\big(|X|^n\big) < \infty$, then writing

$$\mathbb{E}\big(|X|^{n-1}\big) = \mathbb{E}\big(|X|^{n-1}\mathbf{1}_{|X|<1} + |X|^{n-1}\mathbf{1}_{|X|>1}\big) \leq \mathbb{E}\big(\mathbf{1}_{|X|<1} + |X|^n\mathbf{1}_{|X|>1}\big) \leq 1 + \mathbb{E}\big(|X|^n\big)$$

we see $\mathbb{E}\big(|X|^{n-1}\big) < \infty$ also. However the other direction is easily seen not to hold: for every n there is a counterexample X.

Exercise 3. *Given a random variable X such that $\mathbb{E}(X^2) < \infty$ on $(\Omega, \mathcal{F}, \mathbb{P})$, suppose $\mathcal{G} \subseteq \mathcal{F}$ is a σ-algebra.*

Prove that a random variable Y which is $\mathcal{G}$-measurable and which, among $\mathcal{G}$-measurable random variables, minimises $\mathbb{E}\big((Y-X)^2\big)$, is a version of the conditional expectation $\mathbb{E}(X|\mathcal{G})$. You may assume that a minimiser exists.

Solution to Exercise 3:

Let $d = \inf_Y \mathbb{E}\big((Y - X)^2\big)$, where Y ranges over $\mathcal{G}$-measurable random variables. This infimum exists and is a real number because the set is non-empty (it contains $\mathbb{E}(X^2)$, taking Y to be the constant 0 random variable), and bounded below by 0. Suppose Y is a minimiser, i.e. $\mathbb{E}\big((X - Y)^2\big) = d$. By definition we have (ii). Because $\mathbb{E}(X^2) < \infty$ we have $\mathbb{E}(|X|) < \infty$,

and because $\mathbb{E}((Y-X)^2) = d < \infty$, we have $\mathbb{E}(|Y-X|) < \infty$. By the triangle inequality we conclude $\mathbb{E}(|Y|) < \infty$, giving (i).

Suppose there is some $A \in \mathcal{G}$ such that $\mathbb{E}(X\mathbf{1}_A) \neq \mathbb{E}(Y\mathbf{1}A)$. Without loss of generality, suppose $\mathbb{E}(X\mathbf{1}_A) = \mathbb{E}(Y\mathbf{1}A) + \delta$ for some $\delta > 0$. Necessarily we have $\mathbb{P}(A) > 0$, and set $\rho = \delta\mathbb{P}(A)^{-1}$. Then we have

$$\begin{aligned}
\mathbb{E}\big((X-Y-\rho\mathbf{1}_A)^2\big) &= \mathbb{E}\big((X-Y-\rho)^2\mathbf{1}_A\big) + \mathbb{E}\big((X-Y)^2\mathbf{1}_{A^c}\big) \\
&= \mathbb{E}\big((X-Y)^2\mathbf{1}_A - 2\rho(X-Y)\mathbf{1}_A + \rho^2\mathbf{1}_A\big) + \mathbb{E}\big((X-Y)^2\mathbf{1}_{A^c}\big) \\
&= \mathbb{E}\big((X-Y)^2\big) - 2\rho\mathbb{E}\big((X-Y)\mathbf{1}_A\big) + \rho^2\mathbb{P}(A) \\
&= d - 2\rho^2\mathbb{P}(A) + \rho^2\mathbb{P}(A) < d\,,
\end{aligned}$$

which is a contradiction, since $Y + \rho\mathbf{1}_A$ is $\mathcal{G}$-measurable. This verifies (iii); Y is a version of $\mathbb{E}(X|\mathcal{G})$.

9.1 Properties of conditional expectation

Theorem 5. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We assume all random variables X have $\mathbb{E}\big(|X|\big) < \infty$, and we assume all conditional expectations exist. Let $\mathcal{G}$ and $\mathcal{G}'$ be σ-algebras contained in $\mathcal{F}$. Then the following hold.*

(CE1) If X is $\mathcal{G}$-measurable, then $\mathbb{E}(X|\mathcal{G}) = X$ almost surely.

(CE2) For $\alpha, \beta \in \mathbb{R}$ we have $\mathbb{E}(\alpha X + \beta Y|\mathcal{G}) = \alpha\mathbb{E}(X|\mathcal{G}) + \beta\mathbb{E}(Y|\mathcal{G})$ almost surely.

(CE3) If $X \leq Y$ pointwise almost surely, then $\mathbb{E}(X|\mathcal{G}) \leq \mathbb{E}(Y|\mathcal{G})$ pointwise almost surely.

(CE4) If $0 \leq X_1 \leq X_2 \leq \dots$ holds almost surely and almost surely the sequence converges pointwise to X, then $\mathbb{E}(X|\mathcal{G}) = \lim_{n\to\infty}\mathbb{E}(X_n|\mathcal{G})$ almost surely.

(CE5) If almost surely $|X_n| \leq Z$ for all $n \in \mathbb{N}$ and some Z with $\mathbb{E}(|Z|) < \infty$, and $\lim_{n\to\infty} X_n = X$ pointwise almost surely, then $\mathbb{E}(X|\mathcal{G}) = \lim_{n\to\infty}\mathbb{E}(X_n|\mathcal{G})$ almost surely.

(CE6) If $X_1, X_2, \dots \geq 0$ almost surely, then $\mathbb{E}(\liminf_{n\to\infty} X_n|\mathcal{G}) \leq \liminf_{n\to\infty}\mathbb{E}(X_n|\mathcal{G})$ almost surely.

(CE7) If $|X_n| \leq Z$ almost surely for all $n \in \mathbb{N}$ and some Z with $\mathbb{E}(|Z|) < \infty$, then $\mathbb{E}(\limsup_{n\to\infty} X_n|\mathcal{G}) \geq \limsup_{n\to\infty}\mathbb{E}(X_n|\mathcal{G})$ almost surely.

(CE8) Let $g : \mathbb{R} \to \mathbb{R}$ be convex. Then $\mathbb{E}\big(g(X)|\mathcal{G}\big) \geq g\big(\mathbb{E}(X|\mathcal{G})\big)$ almost surely. Furthermore, if $X \in [a,b]$ and $\mathbb{E}(X|\mathcal{G}) = \Lambda a + (1-\Lambda)b$, then $\mathbb{E}\big(g(X)|\mathcal{G}\big) \leq \Lambda g(a) + (1-\Lambda)g(b)$ almost surely. Note Λ is a random variable.

(CE9) If $\mathcal{G}' \subseteq \mathcal{G}$ then $\mathbb{E}\big(\mathbb{E}(X|\mathcal{G})|\mathcal{G}'\big) = \mathbb{E}(X|\mathcal{G}')$ almost surely.

(CE10) If Y is $\mathcal{G}$-measurable and $\mathbb{E}(|XY|) < \infty$, then $\mathbb{E}(XY|\mathcal{G}) = \mathbb{E}(X|\mathcal{G})Y$ almost surely.

(CE11) If $\sigma(X)$ and $\mathcal{G}$ are independent, then $\mathbb{E}(X|\mathcal{G}) = \mathbb{E}(X)$ almost surely.

It is important to recall that 'almost surely' depends on the measure $\mathbb{P}$. In all of these, when we write a conditional expectation in a formula, what is meant is that the statement holds for any version of the conditional expectation. We should also stress that we have not yet proved all the claimed conditional expectations actually exist: at present, we can only apply Theorem 5 where we know the relevant conditional expectations do exist.

Most of these statements are basically from the definitions, plus taking a bit of care to handle the measure zero sets from 'almost surely'. We give the proofs for completeness (i.e. you ought to be able to figure them out easily enough!). Most of them are simply 'conditional versions' of statements we saw for the expectation. However, it's important to remember that while expectations are numbers, conditional expectations are random variables. For (CE8) this matters: the proof is noticeably less easy than the expectation proof.

One thing we need is to explain why we can slightly weaken the assumption of the Monotone Convergence Theorem. If $X_1, \dots$ is a sequence of almost surely non-negative random variables which are almost surely increasing and almost surely converge pointwise to X, we still get $\lim_{n\to\infty} \mathbb{E}(X_n) = \mathbb{E}\big(\lim_{n\to\infty} X_n\big)$.

To see this is true, let for each n the set A_n be the collection of ω where non-negativity or (if $n \geq 2$) $X_n \geq X_{n-1}$ fails; similarly let A be the set where pointwise convergence fails. By assumption, these are all sets of measure zero. We let X'_n for each n, and X', be defined equal to the corresponding X_n or X outside of $A \cup A_1 \cup \dots$ (which is a set of measure zero), and 0 within it. Then we have

$$\lim_{n\to\infty} \mathbb{E}(X_n) = \lim_{n\to\infty} \mathbb{E}(X'_n) = \mathbb{E}(X') = \mathbb{E}(X)$$

as required. The middle equality is the Monotone Convergence Theorem, and the first and last equalities use the fact that setting a function to 0 (or any constant) on a set of measure zero does not change the integral. This last follows easily from the definition of the integral.

By an almost identical argument, we can weaken the assumptions of the Dominated Convergence Theorem.

Proof. For all the formulae we have to prove are conditional expectations, conditions (i) and (ii) are trivially satisfied (the arithmetic and analytic operations on $\mathcal{G}$-measurable functions give us $\mathcal{G}$)-measurable functions), and we will just focus on (iii). Because any two versions of a conditional expectation are almost surely equal (Exercise 2) it is enough to verify that the formula satisfies the definition of a conditional expectation.

For (CE1), condition (ii) is definitional: it requires $\mathbb{E}(X\mathbf{1}_A) = \mathbb{E}(X\mathbf{1}_A)$ for all $A \in \mathcal{G}$.

For (CE2), given $A \in \mathcal{G}$ we have

$$\begin{aligned}\mathbb{E}\big((\alpha\mathbb{E}(X|\mathcal{G}) + \beta\mathbb{E}(Y|\mathcal{G}))\mathbf{1}_A\big) = \alpha\mathbb{E}\big(\mathbb{E}(X|\mathcal{G})\mathbf{1}_A\big) + \beta\mathbb{E}\big(\mathbb{E}(Y|\mathcal{G})\mathbf{1}_A\big) &= \alpha\mathbb{E}(X\mathbf{1}_A) + \beta\mathbb{E}(Y\mathbf{1}_A) \\ &= \mathbb{E}\big((\alpha X + \beta Y)\mathbf{1}_A\big)\end{aligned}$$

as required.

For (CE3), let Z be a version of $\mathbb{E}(X|\mathcal{G})$ and Z' a version of $\mathbb{E}(Y|\mathcal{G})$. Consider the set $A_n = \{\omega : Z(\omega) \leq Z'(\omega) - \frac{1}{n}\}$. This is in $\mathcal{G}$ since $Z - Z'$ is $\mathcal{G}$-measurable, so we have

$$\mathbb{E}\big(X\mathbf{1}_{A_n}\big) = \mathbb{E}\big(Z\mathbf{1}_{A_n}\big) \leq \mathbb{E}\big(Z'\mathbf{1}_{A_n}\big) - \tfrac{1}{n}\mathbb{P}(A_n) = \mathbb{E}\big(Y\mathbf{1}_{A_n}\big) - \tfrac{1}{n}\mathbb{P}(A_n) \leq \mathbb{E}\big(X\mathbf{1}_{A_n}\big) - \tfrac{1}{n}\mathbb{P}(A_n)$$

where the equalities are by (iii) and the inequalities by linearity and monotonicity of the expectation (for the final inequality, we only have almost surely $X \geq Y$, but this still implies $\mathbb{E}(X\mathbf{1}_A) \leq \mathbb{E}(Y\mathbf{1}_A)$ by considering setting $X = Y = 0$ on the set where $X < Y$). We conclude $\mathbb{P}(A_n) = 0$, so by countable additivity $\mathbb{P}\big(\bigcup_{n\in\mathbb{N}} A_n\big) = 0$. Observe that if $\omega \in \Omega$ is such that $Z(\omega) < Z'(\omega)$ then ω is in the union: this proves that almost surely $Z \geq Z'$ as required.

For (CE4), given $A \in \mathcal{G}$ we have

$$\mathbb{E}\Big(\big(\lim_{n\to\infty} \mathbb{E}(X_n|\mathcal{G})\big)\mathbf{1}_A\Big) = \mathbb{E}\Big(\lim_{n\to\infty} \big(\mathbb{E}(X_n|\mathcal{G})\mathbf{1}_A\big)\Big) = \lim_{n\to\infty} \mathbb{E}\big(\mathbb{E}(X_n|\mathcal{G})\mathbf{1}_A\big)\,.$$

Here the final equality is the Monotone Convergence Theorem. To justify this, we need to explain why the sequence of functions $\big(\mathbb{E}(X_n|\mathcal{G})\big)_{n\in\mathbb{N}}$ is almost surely monotone increasing: the reason is (CE3). Continuing, we have

$$\lim_{n\to\infty} \mathbb{E}\big(\mathbb{E}(X_n|\mathcal{G})\mathbf{1}_A\big) = \lim_{n\to\infty} \mathbb{E}\big(X_n\mathbf{1}_A\big) = \mathbb{E}(X\mathbf{1}_A)$$

where the final equality is again the Monotone Convergence Theorem.

Coming to (CE5), we apply much the same logic as for (CE4): we take the limit out of the expectation using the Dominated Convergence Theorem, which is justified because $\big|\mathbb{E}(X_n|\mathcal{G})\big|$ is pointwise almost surely bounded above by $\mathbb{E}(Z|\mathcal{G})$ by (CE3). We then get the same chain of logic:

$$\begin{aligned}\mathbb{E}\Big(\big(\lim_{n\to\infty} \mathbb{E}(X_n|\mathcal{G})\big)\mathbf{1}_A\Big) &= \mathbb{E}\Big(\lim_{n\to\infty} \big(\mathbb{E}(X_n|\mathcal{G})\mathbf{1}_A\big)\Big) = \lim_{n\to\infty} \mathbb{E}\big(\mathbb{E}(X_n|\mathcal{G})\mathbf{1}_A\big)\\ &= \lim_{n\to\infty} \mathbb{E}\big(X_n\mathbf{1}_A\big) = \mathbb{E}(X\mathbf{1}_A)\,,\end{aligned}$$

where this time the second and final equalities are the Dominated Convergence Theorem.

The proof of (CE6) is much the same as the proof of Fatou's Lemma. We have

$$\begin{aligned}\mathbb{E}\big(\liminf_{n\to\infty} X_n|\mathcal{G}\big) &= \mathbb{E}\big(\lim_{n\to\infty} \inf_{k\geq n} X_k|\mathcal{G}\big)\\ &= \lim_{n\to\infty} \mathbb{E}\big(\inf_{k\geq n} X_k|\mathcal{G}\big)\\ &\leq \lim_{n\to\infty} \inf_{k\geq n} \mathbb{E}\big(X_k|\mathcal{G}\big)\\ &= \liminf_{n\to\infty} \mathbb{E}\big(X_n|\mathcal{G}\big)\,,\end{aligned}$$

where all these lines hold almost surely: the first equality is by definition of limit infimum, the second is (CE4), which applies because the sequence $\inf_{k\geq n} x_k$ is (surely) monotone increasing, and the inequality is (CE3), with the final equality again the definition of limit infimum. For the inequality, again a little care is needed: (CE3) tells us $\mathbb{E}(X_t|\mathcal{G}) \geq \mathbb{E}\big(\inf_{k\geq n} X_k|\mathcal{G}\big)$ holds for all $t \geq n$ almost surely; to conclude the same for the infimum over $t \geq n$ requires observing that the union of the corresponding sets of measure zero where the inequality fails is a set of measure zero.

Proving (CE7) is done as in the proof of the reverse Fatou inequality from Notes 2.

Passing over (CE8) briefly, (CE9) (the 'tower property') follows since

$$\mathbb{E}\Big(\mathbb{E}\big(\mathbb{E}(X|\mathcal{G})|\mathcal{G}'\big)\mathbf{1}_A\Big) = \mathbb{E}\big(\mathbb{E}(X|\mathcal{G})\mathbf{1}_A\big) = \mathbb{E}(X\mathbf{1}A)\,,$$

For (CE10) ('taking out what is known', because Y is known if we condition on $\mathcal{G}$), suppose first $Y \geq 0$. Let $Y' = \sum_i \alpha_i \mathbf{1}_{B_i}$ be a $\mathcal{G}$-measurable simple function bounded between 0 and Y pointwise, and we have for any $A \in \mathcal{G}$:

$$\mathbb{E}\Big(\mathbb{E}(X|\mathcal{G})Y'\mathbf{1}_A\Big) = \sum_i \alpha_i \mathbb{E}\Big(\mathbb{E}(X|\mathcal{G})\mathbf{1}_{A\cap B_i}\Big) = \sum_i \alpha_i \mathbb{E}\big(X\mathbf{1}_{A\cap B_i}\big) = \mathbb{E}(XY'\mathbf{1}_A)\,,$$

where the first equality is (CE2), the second is condition (iii) and the final one is linearity of expectation (and $\mathbf{1}_{A\cap B_i} = \mathbf{1}_A\mathbf{1}_{B_i}$). Now taking a monotone increasing sequence of $\mathcal{G}$-measurable simple functions converging to Y, and applying the Monotone Convergence Theorem twice, we get

$$\mathbb{E}\Big(\mathbb{E}(X|\mathcal{G})Y\mathbf{1}_A\Big) = \mathbb{E}(XY\mathbf{1}_A)$$

as required. Finally, for (CE11), if $\sigma(X)$ and $\mathcal{G}$ are independent, we have

$$\mathbb{E}\big(\mathbb{E}(X|\mathcal{G})\mathbf{1}_A\big) = \mathbb{E}(X\mathbf{1}_A) = \mathbb{E}(X)\mathbb{E}(\mathbf{1}_A) = \mathbb{E}\big(\mathbb{E}(X)\mathbf{1}_A\big)$$

where we used multiplicativity of the expectation for the independent random variables X and $\mathbf{1}_A$.

Finally, we return to (CE8). The conditional reverse Jensen inequality is easy: by convexity if $x = \lambda a + (1-\lambda)b$ then $g(x) \leq \lambda g(a) + (1-\lambda)g(b)$. We have $\lambda = \frac{b-x}{b-a}$, so we can rewrite this as

$$g(x) \leq \tfrac{b-x}{b-a} + \tfrac{x-a}{b-a}g(b)$$

and now (CE3) says

$$\mathbb{E}\big(g(X)|\mathcal{G}\big) \leq \mathbb{E}\Big(\tfrac{b-X}{b-a} + \tfrac{X-a}{b-a}g(b)|\mathcal{G}\Big) = \frac{b-\mathbb{E}(X|\mathcal{G})}{b-a} + \frac{\mathbb{E}(X|\mathcal{G})-a}{b-a}$$

almost surely. Defining $\Lambda = \frac{b-\mathbb{E}(X|\mathcal{G})}{b-a}$ and substituting in we obtain the required inequality.

What is a bit more difficult is the conditional Jensen inequality. If $g(x) = ax + b$ is linear, we can use the same argument as above with the inequality reversed. However if not, we cannot fix a single linear approximation ('subgradient') at the 'right place': that would require us to know $\mathbb{E}(X|\mathcal{G})$, but that is not a number (as $\mathbb{E}(X)$ is in the proof of Jensen's inequality) but itself a random variable.

Instead, consider the set L of all lines $px + q$ such that $p, q \in \mathbb{Q}$ and $px + q \leq g(x)$ holds for all x. We Claim that this set is non-empty and for any (x, y) with $y < g(x)$ there exists a line in L which passes above (x, y). Assuming for a moment this claim holds, we have

$$\mathbb{E}\big(g(X)|\mathcal{G}\big) \geq \mathbb{E}\big(pX + q|\mathcal{G}\big) = p\mathbb{E}(X|\mathcal{G}) + q$$

where we fix a version of the left and right conditional expectations throughout, by (CE3) and (CE2) almost surely, for any $px + q \in L$. For each $\ell \in L$, let A_ℓ be the set on which the above fails, which is of measure zero. Since $|L| \leq |\mathbb{Q}|^2$ is countable, letting $A = \bigcup_{\ell\in L}$ we see that A has measure zero.

Given $\omega \notin A$, let $y = \mathbb{E}\big(g(X)|\mathcal{G}\big)(\omega)$ and $x = \mathbb{E}(X|\mathcal{G})(x)$. If $y < g(x)$, then there exists $\ell \in L$ such that $px + q > y$ by our claim, which contradicts the definition of A. So $y \geq g(x)$ as required.

What remains here is to prove the Claim. This is a fairly straightforward but annoying piece of real analysis; we give the ideas. We needed to exclude the case g is linear because a line with irrational gradient would contradict the Claim. Now given any n sufficiently large that there are subgradients to g at $-n$ and at n which are different (this exists because g is not linear), for any larger n the same property remains true. Convexity implies that the value of any subgradient at c is increasing in c (if this fails, draw subtangent lines at c and $c' > c$ with decreasing slope: these lines form a triangle with c, c' and their intersection above the line from $(c, g(c))$ to $(c', g(c'))$; by convexity the function lies below this line but by definition of a subtangent the function lies above the triangle, a contradiction), so picking a subtangent $m(x-c)+g(c)$ at any given $c \in (-n,n) \cap \mathbb{Q}$, for any rational $q < g(c)$ we can choose a rational approximation p to m which is between the subgradients at $-n$ and at n, and such that $px+q$ lies below $m(x-c)+g(c)$ on $[-n,n]$. Then this line lies below $g(x)$ on $(-\infty,\infty)$ and (because the rationals are dense in the reals) this set of lines separates any (x,y) with $y < g(x)$ from $g(x)$, not just at rational x. $\square$

We should stress here that we did a lot of repeating essentially proofs we saw before, or quoting results for the expectation, plus being a bit careful to justify statements about 'almost surely'. Some authors expect the reader to fill in the missing details on this, sometimes even without pointing out that the details are there—be careful, because otherwise you'll fall in to the trap of trying to prove (CE8) by considering the set of all lines below the curve (not just those with rational coefficients). This looks like it avoids the need to do linear g as a special case, or to bother proving the Claim: but then you end up trying to take the union of an uncountable collection of measure zero sets!

Exercise 4. *Suppose X and Y are simple random variables with $\mathbb{E}(|X|), \mathbb{E}(|Y|) < \infty$, and write $X = \sum_i \alpha_i \mathbf{1}_{A_i}$ and $Y = \sum_i \beta_i \mathbf{1}_{B_i}$; suppose (without loss of generality) that the α_i are distinct and the A_i pairwise disjoint, and similarly for the β_j and B_j. Define the* conditional probability mass function of X given Y *by*

$$p_{X|Y}(x|y) = \begin{cases} \frac{\mathbb{P}(X=x, Y=y)}{\mathbb{P}(Y=y)} & \text{if } \mathbb{P}(Y=y) > 0 \\ 0 & \text{if } \mathbb{P}(Y=y) = 0 \end{cases} .$$

Prove that

$$Z = \sum_i \Big(\sum_j x_j p_{X|Y}(x_j|y_i) \Big) \mathbf{1}_{B_i}$$

is a version of $\mathbb{E}(X|Y)$.

Solution to Exercise 4:

Here measurability (ii) is trivial: each B_i is in $\sigma(Y)$ by definition.

For (i), we calculate $\mathbb{E}(|Z|)$ directly, as a finite sum:

$$\mathbb{E}(|Z|) = \sum_i \Big| \sum_j x_j p_{X|Y}(x_j|y_i) \Big| \mathbb{P}(B_i) \leq \sum_i \sum_j |x_j| p_{X|Y}(x_j|y_i) \mathbb{P}(B_i) = \sum_j |x_j| \mathbb{P}(X = x_j) = \mathbb{E}(|X|) ,$$

where we are allowed to swap the order of finite sums, and we use the fact $\sum_i \mathbb{P}(X = x_j, Y = y_i) = \mathbb{P}(X = x_j)$.

Finally, for (iii), we do a similar calculation. Given $C \in \sigma(Y)$, recall $\sigma(Y)$ is a finite set and hence by linearity of expectation, it will be enough to prove $\mathbb{E}(X\mathbf{1}_C) = \mathbb{E}(Z\mathbf{1}_C)$ for a

minimal non-empty set C in $\sigma(Y)$. We may further assume (because the statement is trivial otherwise) that $\mathbb{P}(C) > 0$.

Then
$$\mathbb{E}(Y\mathbf{1}_C) = \sum_i \Big(\sum_j x_j p_{X|Y}(x_j|y_i) \Big) \mathbb{P}\big(B_i \cap C\big) .$$

Because $B_i \in \sigma(Y)$ and by disjointness and minimality of C we have either $B_i = C$ or $B_i \cap C = \emptyset$; again by disjointness there is at most one value of i such that the former holds. Again without loss of generality (adding if necessary an indicator function with coefficient zero) we may assume there is exactly one i. For this fixed i, because the event $Y = y_i$ is C, we get

$$\mathbb{E}(Y\mathbf{1}_C) = \sum_j x_j p(X|Y)(x_j|y_i)\mathbb{P}(C) = \sum_j x_j \frac{\mathbb{P}(X = x_j, Y = y_i)}{\mathbb{P}(Y = y_i)} \mathbb{P}(C) = \sum_j x_j \mathbb{E}(\mathbf{1}_{A_j}\mathbf{1}_C) = \mathbb{E}(X\mathbf{1}_C)$$

as required.

Given a pair of random variables X and Y, the *joint probability density function* is $f_{XY}(x, y)$ a non-negative function on $\mathbb{R}^2$ such that

$$\mathbb{P}\big((X, Y) \in A\big) = \int_A f_{XY}(x, y)\, \mathrm{d}(x, y) ,$$

for all Borel $A \subseteq \mathbb{R}^2$. Note that this does not always exist; when it does we say X and Y are *jointly continuous*. It is easy to check (using Fubini's Theorem!) that if X and Y are both continuous (so have pdfs f_X and f_Y) and independent, then $f_{XY}(x, y) = f_X(x)f_Y(y)$ does exist (and conversely if $f_{XY}(x, y) = f_X(x)f_Y(y)$ then X and Y are independent); the 'interesting case' is however when we don't have independence.

Exercise 5. *Suppose X and Y are jointly continuous random variables with $\mathbb{E}(|X|), \mathbb{E}(|Y|) < \infty$. Define the* conditional probability density function of X with respect to Y *to be*

$$f_{X|Y}(x|y) = \begin{cases} \frac{f_{XY}(x,y)}{f_Y(y)} & \text{if } f_Y(y) \neq 0 \\ 0 & \text{if } f_Y(y) = 0 \end{cases} .$$

Prove that $\int_{x=-\infty}^{\infty} x f_{X|Y}(x, Y)\, \mathrm{d}x$ is a version of $\mathbb{E}(X|Y)$.

Solution to Exercise 5:

This integral can be interpreted to make sense: since both sides are functions of $\omega \in \Omega$, we really have a whole collection of integrals, one for each ω. We do however need to be a bit careful: f_{XY} is a measurable function on $\mathbb{R}^2$, but this does not imply that for all fixed y the function $x \to f_{XY}(x, y)$ is measurable, and indeed that might fail on a set of measure zero. Fortunately, because the conditional expectation is only defined up to almost sure equality, we can safely take the integral to be equal to zero for ω such that $f_{XY}(x, Y(\omega))$ is not a measurable function of x. (We should really justify the statement that non-measurability can only occur on a set of measure zero; we should also have proved Fubini's Theorem, which we are about to use and which needs the same justification: but there was not time in the course.) In the following calculation, we should formally be somewhat careful too about writing e.g. $f_X(x)$ as though it is a single function, when really it's defined up to almost sure

equality. Furthermore, formally $\int_{y=-\infty}^{\infty} f_{XY}(x,y)\,\mathrm{d}y$ might not be defined for some values of x; but the values will be a set of measure zero and, because we're integrating over x, we can ignore the problem. The 'nicest' way to make this be formally correct is simply to define the Lebesgue integral for all non-negative functions (not just measurable ones) by the supremum definition we gave, and then to say that we only obtain the various nice properties of the integral for measurable functions.

We prove that the given integral $Z = \int_{x=-\infty}^{\infty} x f_{X|Y}(x,Y)\,\mathrm{d}x$ satisfies the properties of conditional expectation. For the two integration statements (i) and (iii), we use the LOTUS and Fubini's theorem. For (i), we have

$$\begin{aligned}
\mathbb{E}(|Z|) &= \int_\Omega \Big| \int_{x=-\infty}^{\infty} x f_{X|Y}(x,Y)\,\mathrm{d}x \Big| \,\mathrm{d}\mathbb{P} \\
&\le \int_\Omega \int_{x=-\infty}^{\infty} |x| f_{X|Y}(x,Y)\,\mathrm{d}x\,\mathrm{d}\mathbb{P} \\
&= \int_{x=-\infty}^{\infty} \int_\Omega |x| f_{X|Y}(x,Y)\,\mathrm{d}\mathbb{P}\,\mathrm{d}x \\
&= \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} |x| f_{X|Y}(x,y) f_Y(y)\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_{x=-\infty}^{\infty} |x| \int_{y=-\infty}^{\infty} f_{XY}(x,y)\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_{x=-\infty}^{\infty} |x| f_X(x)\,\mathrm{d}x \\
&= \mathbb{E}(|X|)\,,
\end{aligned}$$

where we used monotonicity to take the absolute value inside the integral, Fubini's theorem to swap the integrals, and the LOTUS twice to go between integrating $\mathrm{d}\mathbb{P}$ and integrating over the reals.

The calculation for (iii) is quite similar. Given $A \in \sigma(Y)$, we have

$$\begin{aligned}
\mathbb{E}(\mathbf{1}_A Z) &= \int_\Omega \mathbf{1}_A \int_{x=-\infty}^{\infty} x f_{X|Y}(x,Y)\,\mathrm{d}x\,\mathrm{d}\mathbb{P} \\
&= \int_\Omega \int_{x=-\infty}^{\infty} x f_{X|Y}(x,Y(\omega))\mathbf{1}_A(\omega)\,\mathrm{d}x\,\mathrm{d}\mathbb{P} \\
&= \int_{x=-\infty}^{\infty} \int_\Omega x f_{X|Y}(x,Y(\omega))\mathbf{1}_A(\omega)\,\mathrm{d}\mathbb{P}\,\mathrm{d}x \\
&= \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} x f_{X|Y}(x,y)\mathbf{1}_{y\in Y(A)} f_Y(y)\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{\infty} x f_{XY}(x,y)\mathbf{1}_{y\in Y(A)}\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_{(x,y)\in\mathbb{R}^2} x f_{XY}(x,y)\mathbf{1}_{y\in Y(A)}\,\mathrm{d}\lambda(x,y)
\end{aligned}$$

This integral is equal—by a two-dimensional version of the LOTUS, whose proof is basically the same as the one-dimensional version—to $\mathbb{E}_{X,Y}(X\mathbf{1}_{Y\in Y(A)}) = \mathbb{E}_{X,Y}(X\mathbf{1}_A) = \mathbb{E}(X\mathbf{1}A)$ as required. Note here we are using the fact $A \in \sigma(Y)$ to conclude that $Y(\omega) \in Y(A)$

if and only if $\omega \in A$. The backwards implication is true for any function (by definition). For the forwards implication, we know that $A = Y^{-1}(B)$ for some Borel set B, because $A \in \sigma(Y)$. Certainly B contains $Y(A)$ (if $Y(\omega) \notin B$ for some $\omega \in A$, by definition this ω is not in $Y^{-1}(B)$); but then a failure of the forwards implication means some $\omega \notin A$ has $Y(\omega) \in Y(A) \subseteq B$, but then $\omega \in Y^{-1}(B)$.

Finally, for $\sigma(Y)$-measurability, we use the fact (which we did not prove!) that $g(y) = \int_{x=-\infty}^{\infty} f_{X|Y}(x, y)\,\mathrm{d}x$ is a measurable function of y. Then $g(Y)$ is a $\sigma(Y)$-measurable function as proved in Notes 2.

9.2 Radon-Nikodym Theorem

In this section we will deal with a measurable space $(X, \mathcal{F})$ with two different measures μ, ν on it. Recall that μ is called σ-finite if we can write X as a countable union of measurable sets with finite μ-measure.

The 'immediate' reason for this is that it will be useful in order to prove conditional expectations exist. However, this is actually a very useful concept in general.

In probability, we often want to analyse some 'natural' probability measure $\mathbb{P}$, such as the uniform distribution on a set. For an example where this is practically interesting, take the collection of 1-Lipschitz functions on the square grid $\{0, 1, \ldots, n\}^2$: that is, the set of functions f such that $|f(a, b) - f(a', b')| \leq 1$ whenever either $a = a'$ and $|b - b'| = 1$ or $|a - a'| = 1$ and $b = b'$, with the condition $f(0, 0) = 0$. In statistical mechanics, it's interesting to study the uniform distribution on these functions $\mathbb{P}$.

Unfortunately, $\mathbb{P}$ is very hard to work with: function values at even widely separated vertices are correlated, and there is no obvious way to understand the correlation. In fact, there are (still open) conjectures that say $\mathbb{P}$ looks very different to, for example, a product measure. However, some simple questions can be answered by finding a different, 'easier' measure $\mathbb{Q}$ on the same set, proving that (for example) a certain property is extremely unlikely with respect to $\mathbb{Q}$, and then arguing that $\mathbb{P}$ and $\mathbb{Q}$ are 'not too different'.

One example of a measure $\mathbb{Q}$ we might consider is the following: we construct a random function f by going along rows of the grid one after another starting from $(1, 0)$ (since $(0, 0)$ is fixed). On the first row, at each $(a, 0)$ choose a uniform random number $f((a, 0))$ in $[f((a-1, 0)) - 1, f((a-1, 0)) + 1]$. In each subsequent row, choose $f((0, b))$ uniformly in $[f((0, b-1)) - 1, f((0, b-1)) + 1]$, and then for $a >$, choose $f(a, b)$ uniformly in the intersection $[f((a-1, b)) - 1, f((a-1, b)) + 1] \cap [f((a, b-1)) - 1, f((a, b-1)) + 1]$.

It's immediate from the construction that the f we get like this is 1-Lipschitz: every time we chose a value, we ensured it is sufficiently close to the previously chosen values on the neighbours. And the construction cannot 'fail': that is, there is always a set of elements to choose from. This isn't quite obvious for choosing $f(a, b)$ when $a, b > 0$: what if the intersection is empty? But this cannot be, because $f(a-1, b)$ and $f(a, b-1)$ are both within distance 1 of $f(a-1, b-1)$, so their mutual distance is at most 2, and the intersection is non-empty.

This measure is easier to understand than $\mathbb{P}$; looking ahead to the next section, it gives a natural way to make some random variables be a martingale sequence; we've built in a lot of independence. The measure does generate every 1-Lipschitz function: indeed, it's

quite easy, given a 1-Lipschitz function f, to calculate the probability density at f; just look at how big all the relevant intersection sets are. However, it doesn't generate the uniform measure: because the intersection sizes will be different for different 1-Lipschitz f, the probability density function is not going to be constant. Nevertheless, if an event has very small $\mathbb{Q}$-probability, it's also going to have small $\mathbb{P}$-probability.

Giving anything resembling details here would take us too far off course: but this strategy shows up often in proofs.

What we care about for this to work is: can we quantify 'how different' our measures are?

In financial mathematics, the concept also shows up naturally. Since financial mathematics is just gambling, think of placing a bet on the flip of a coin: you bet £x (which can be negative!), and you will get back nothing if the coin comes down Tails, and £$2x$ if it comes down Heads (If your bet was negative, that means you must pay up!). Or you can just put your money in the bank and receive £$(1+r)x$ for every saved pound. To put it another way, $B_0 = 1$ is the cost of a bet, and B_1 is uniformly distributed on $\{0, 2\}$ is the return of the bet.

For this setup to make any sense, we want $0 < r < 1$. If $r \geq 1$, you'd bet arbitrarily heavily against the coin, i.e. take x very large and negative, and put the resulting money in the back: even if the coin comes up Heads, the interest pays what you need to pay up and (if $r > 1$) even some more.

In this example, $\mathbb{P}$ is the 'obvious' probability measure: Heads and Tails are equally likely.

However, there is a 'useful' different measure we can consider: we can try to take account of the possibility of betting against the coin and banking the money we get. If we define $q = \frac{1+r}{2}$, we can consider the measure $\mathbb{Q}$ in which Heads is assigned measure q and Tails $1-q$. This is a probability measure (by definition), but of course it's not actually 'the probability' of Heads or Tails, just a convenient way of assigning relative value. The point is that the expected value of a £1 bet in this measure is $\mathbb{E}_{\mathbb{Q}} B_1 = 1 + r$, the same as if you chose not to bet at all but to bank the money.

A ~~fraudster~~ trader now offers you a financial product: if you buy it, they will pay you £a if the coin comes down Tails, and £b if it comes down Heads. Suppose $a > 0$. How much is the product worth?

The right answer is certainly not $\frac{a+b}{2}$ (which is just the expectation of the payment): if you buy the trader's product at that price, they can take your money, place a bet £$\frac{b-a}{2}$ on the coin, and put the remaining £a in the bank. If the coin comes down Tails, they lose their money, but they can pay your £a and still keep the interest £ra; if the coin comes down Heads, they win their bet, so they have in total $2\frac{b-a}{2} + (1+r)a = b + ar$; again, this is enough to make the payout £b and still keep a profit £ra. This is free money for the trader, and the money came from you overpaying.

In this example, it's easy enough to analyse by hand and find that the 'right' price is

$$\frac{qb+(1-q)a}{1+r},$$

which is the expected return under $\mathbb{Q}$ of the trader's product, discounted by the interest rate $1 + r$ (it's natural to do that: because you have the money now and you're comparing

it to the money you'll get after the coin flip; if you'd banked the money instead you'd have $1+r$ times your banked money after the coin flip). The 'point' in this context is that this *risk-neutral measure* is a general way to figure out fair prices.

To make this work, we need to be able to calculate in the risk-neutral measure. One way to do this is to quantify how different it is to the actual measure.

Suppose there is a non-negative measurable $f : X \to \mathbb{R}$ such that

$$\nu(A) = \int_{x \in A} f(x)\,\mathrm{d}\mu = \int_X f\mathbf{1}_A\,\mathrm{d}\mu$$

holds for all $A \in \mathcal{F}$.

Then we have $\mu(A) = 0 \implies \nu(A) = 0$. This is immediate from the definition of the integral: whatever simple g we take bounded above by $f\mathbf{1}_A$, it is bounded above by $\infty\mathbf{1}_A$ which has integral zero. The converse obviously fails: if $f = 0$ on a measurable set A', then we will have $\nu(A') = 0$ whatever $\mu(A')$ is.

If μ, ν are any two measures on $(X, \mathcal{F})$, we say ν *is absolutely continuous with respect to* μ, and write $\nu \ll \mu$, if $\mu(A) = 0 \implies \nu(A) = 0$ for all $A \in \mathcal{F}$. Observe that if a condition holds μ-almost everywhere (i.e. it fails on a set E with $\mu(E) = 0$) then by definition the condition holds ν-almost everywhere; the converse fails.

So what we just did is justify that if we have f such that $\nu(A) = \int_A f\,\mathrm{d}\mu$, then $\nu \ll \mu$. The Radon-Nikodym Theorem is a (partial, but we will only care about σ-finite measures) converse to this.

Theorem 6 (Radon-Nikodym Theorem)**.** *Let μ, ν be σ-finite measures on $(X, \mathcal{F})$. If $\nu \ll \mu$, then there exists $f : X \to \mathbb{R}$ non-negative and $\mathcal{F}$-measurable, such that for all $A \in \mathcal{F}$ we have*

$$\nu(A) = \int_X f\mathbf{1}_A\,\mathrm{d}\mu\,.$$

We call any f which makes the above equation true the *Radon-Nikodym derivative* $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}$. It's immediate that if $f = g$ holds μ-almost everywhere and f is the Radon-Nikodym derivative $\frac{\mathrm{d}\nu}{\mathrm{d}\mu}$, then so is g.

It's also easy to check the converse: if f and g are both the Radon-Nikodym derivative, then letting A_n be the set of x such that $f(x) \geq g(x) + \frac{1}{n}$ we have

$$\nu(A_n) = \int_X f\mathbf{1}_A\,\mathrm{d}\mu \geq \int_X (g + \frac{1}{n})\mathbf{1}_A\,\mathrm{d}\mu = \int_X g\mathbf{1}_A\,\mathrm{d}\mu + \tfrac{1}{n}\mu(A_n) = \nu(A_n) + \tfrac{1}{n}\mu(A_n)$$

so $\mu(A_n) = 0$, so $\mu\big(\bigcup_{A_n}\big) = 0$, so $f > g$ holds on a set of measure zero. By symmetry also $g > f$ holds on a set of measure zero, so $f = g$ holds μ-almost everywhere.

The Radon-Nikodym derivative is defined up to equality μ-almost everywhere.

The Radon-Nikodym derivative quantifies 'how different' μ and ν are. If for example we know it is almost everywhere bounded between $\frac{1}{2}$ and 2, then μ and ν might not be the same measure, but they give the same answer up to a factor of 2 for the measure of any set. (This might well be 'enough for proofs').

More generally, we have the *change of measure* integral formula.

Theorem 7. *Suppose $\nu \ll \mu$ are σ-finite measures on $(X, \mathcal{F})$. Then for any $g : X \to \mathbb{R}$ measurable we have*

$$\int_X g \, \mathrm{d}\nu = \int_X g \frac{\mathrm{d}\nu}{\mathrm{d}\mu} \, \mathrm{d}\mu \, .$$

Proof. As usual, if we can show this holds for any non-negative g, then the result for general g holds by writing $g = g^+ - g^-$ and using linearity of the integral.

Suppose $g \geq 0$, and let $g_1, \dots$ be a non-negative, monotone increasing sequence of simple functions converging to g pointwise. Then we have

$$\begin{aligned}\int_X g \, \mathrm{d}\nu &= \lim_{n\to\infty} \int_X g_n \, \mathrm{d}\nu = \lim_{n\to\infty} \sum_i \alpha_{i,n} \nu(A_{i,n}) \\ &= \lim_{n\to\infty} \sum_i \alpha_{i,n} \int_X \mathbf{1}_{A_{i,n}} \frac{\mathrm{d}\nu}{\mathrm{d}\mu} \, \mathrm{d}\mu = \lim_{n\to\infty} \int_X g_n \frac{\mathrm{d}\nu}{\mathrm{d}\mu} \, \mathrm{d}\mu = \int_X g \frac{\mathrm{d}\nu}{\mathrm{d}\mu} \, \mathrm{d}\mu \, .\end{aligned}$$

where the outside equalities are the Monotone Convergence Theorem. □

This theorem justifies the word 'derivative': the Radon-Nikodym derivative behaves like the usual derivative in a change of measure. In addition, if μ is the Lebesgue measure on $\mathbb{R}$ and ν is any continuous distribution, with cumulative density function F, then $\frac{\mathrm{d}\nu}{\mathrm{d}\mu} = F'$ is a derivative in the usual sense of the word.

9.3 Existence of conditional expectation

Exercise 6. *Let X be a non-negative random variable on $(\Omega, \mathcal{F}, \mathbb{P})$. Define a function $\nu : \mathcal{F} \to \mathbb{R}^\infty_{\geq 0}$ by*

$$\nu(A) = \int_\Omega X \mathbf{1}_A \, \mathrm{d}\mathbb{P} \, .$$

Prove that ν is a measure on $\mathcal{F}$.

Solution to Exercise 6:

Trivially we have $\nu(\emptyset) = 0$ and $\nu \geq 0$, so we need to verify countable additivity. Let $A_1, \dots$ be disjoint sets in $\mathcal{F}$. Then we have

$$\nu\Big(\bigcup_{n\in\mathbb{N}} A_n\Big) = \int_\Omega X \sum_{n=1}^\infty \mathbf{1}_{A_n} \, \mathrm{d}\mathbb{P} = \lim_{n\to\infty} \int_\Omega X \sum_{k=1}^n \mathbf{1}_{A_k} \, \mathrm{d}\mathbb{P} = \lim_{n\to\infty} \sum_{k=1}^n \nu(A_k) = \sum_{n=1}^\infty \nu(A_n)$$

as required. Here the first equality is the definition of ν and the fact that the A_n are disjoint; the second is the Monotone Convergence Theorem; the third uses linearity of the integral to pull the finite sum out, and then the definition of ν, and the final equality is the definition of the series.

Theorem 8. *Let X be a random variable on $(\Omega, \mathcal{F}, \mathbb{P})$, and suppose $\mathbb{E}(|X|) < \infty$. Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ-algebra. Then $\mathbb{E}(X|\mathcal{G})$ exists.*

Proof. By the usual trick, it suffices to prove the theorem for $X \geq 0$: assuming we have this, then $\mathbb{E}(X^+|\mathcal{G})$ and $\mathbb{E}(X^-|\mathcal{G})$ exist, and their difference was proved in Theorem 5 to be a version of $\mathbb{E}(X|\mathcal{G})$.

Suppose $X \geq 0$. We define a measure ν (by Exercise 6) on $(\Omega, \mathcal{G})$ by

$$\nu(A) = \int_\Omega X \mathbf{1}_A \, \mathrm{d}\mathbb{P},$$

and we let μ be the measure on $(\Omega, \mathcal{G})$ obtained by $\mu(A) = \mathbb{P}(A)$. We claim that $\nu \ll \mu$: indeed, this follows directly from the definition of the integral.

Let $Y = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}$ exists by Theorem 6, applied with these two measures on $(\Omega, \mathcal{G})$.

By definition we have $1 = \nu(\Omega) = \mathbb{E}_\mu\big(Y\big)$ which verifies (i).

Theorem 6 explicitly guarantees that Y is $\mathcal{G}$-measurable, giving (ii).

For (iii), given $A \in \mathcal{G}$ we have

$$\mathbb{E}_\mathbb{P}(Y\mathbf{1}_A) = \int_\Omega \mathbf{1}_A \tfrac{\mathrm{d}\nu}{\mathrm{d}\mu} \, \mathrm{d}\mathbb{P} = \int_\Omega \mathbf{1}_A \tfrac{\mathrm{d}\nu}{\mathrm{d}\mu} \, \mathrm{d}\mu = \int_\Omega \mathbf{1}_A \, \mathrm{d}\nu = \nu(A) = \int_\Omega X\mathbf{1}_A \, \mathrm{d}\mathbb{P} = \mathbb{E}_\mathbb{P}(X\mathbf{1}A)$$

where the middle equality is Theorem 7. To justify the second equality, we can use a general fact: if f is any non-negative $\mathcal{G}$-measurable function, letting $f_1, \ldots$ be a sequence of non-negative simple $\mathcal{G}$-measurable functions which are monotone increasing and converging pointwise to f, we have

$$\int_\Omega f \, \mathrm{d}\mathbb{P} = \lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mathbb{P} = \lim_{n\to\infty} \int_\Omega f_n \, \mathrm{d}\mu = \int_\Omega f \, \mathrm{d}\mu$$

where the outside equalities are the Monotone Convergence Theorem and the middle equality uses the definition of the Lebesgue integral for simple functions and the fact $\mu(B) = \mathbb{P}(B)$ for all $\mathcal{G}$-measurable B.

For clarity, let's give the detail of the middle equality properly. We want to prove $\int_\Omega f_n \, \mathrm{d}\mathbb{P} = \int_\Omega f_n \, \mathrm{d}\mu$. Write $f_n = \sum_i \alpha_i \mathbf{1}_{B_i}$ as a finite sum (because f_n is simple), where each B_i is in $\mathcal{G}$ (which we can because f_n is $\mathcal{G}$-measurable). Then we have

$$\int_\Omega f_n \, \mathrm{d}\mathbb{P} = \sum_i \alpha_i \mathbb{P}(B_i) = \sum_i \alpha_i \mu(B_i) = \int_\Omega f_n \, \mathrm{d}\mu$$

where the outside equalities are as promised the definition of the integral for simple functions, and the middle equality uses $\mathbb{P}(B_i) = \mu(B_i)$ for each i, which holds because $B_i \in \mathcal{G}$ and we defined μ to equal $\mathbb{P}$ on $\mathcal{G}$. $\square$

10 Martingales

In this section, we are thinking about a *discrete-time process*: we start at time 0, and then at each $n \in \mathbb{N}$ (possibly up to some finite future time N) a piece of randomness occurs. We want to understand the result.

The simplest 'piece of randomness' to understand is the following setup: something random occurs, it generates for us a random variable Y_n which has conditional expectation $\mathbb{E}(Y_n|\mathcal{F}_{n-1}) = 0$, and we want to know how $\sum_{n=1}^{N} Y_n$ behaves.

We should be clear what $\mathcal{F}_{n-1}$ is. It is a σ-algebra; one 'minimal' possibility is $\sigma(Y_1, \dots, Y_{n-1})$, but it could be larger. We won't generally need to be too specific: we will want to think of $\mathcal{F}_{n-1}$ as carrying all the information about the random choices up to and including time $n-1$, but not about time n. So Y_n will be $\mathcal{F}_n$-measurable, but not $\mathcal{F}_{n-1}$-measurable. However, $\mathbb{E}(Y_n|\mathcal{F}_{n-1}) = 0$ does *not* tell us that Y_n is independent of $Y_1, \dots, Y_{n-1}$.

One example to think about is that we're placing bets at every time step. Each time step, depending on a fair coin toss at time n, we will either lose or gain £x, where x is a number we are free to choose after seeing the outcomes of all our previous bets. We can fix our favourite rule on how to bet given the outcomes: the conditional expectation is always zero, but the random variables generally aren't independent.

Of course, our random variables don't have to be two-outcome random variables, or even discrete: anything goes. A *discrete-time process* is a sequence $X_0, X_1, \dots$ of random variables. An example of a discrete-time process is $X_n = \sum_{i=1}^{n} Y_i$, and of course we can write any discrete-time process in the summation form ($Y_n = X_n - X_{n-1}$); depending on circumstances, it can be more natural to think of a process or to think of a sum of random variables.

In this course we will not meet any continuous time process (or anything other than what we defined as a discrete-time process) and we will simply write *process* from now on. Even if we are interested in a process which halts at some finite time N, we will formally define the entire infinite sequence: in this case $\mathcal{F}_n = \mathcal{F}_N$ and $X_n = X_N$ for all $n > N$.

Definition 9 (Filtration, process adapted to a filtration, natural filtration)**.** *A* filtration *on* $(\Omega, \mathcal{F}, \mathbb{P})$ *is a sequence* $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \dots$ *of* σ*-algebras contained in* $\mathcal{F}$*; we often write* $(\mathcal{F}_n)$*. We sometimes say* filtered probability space *for a probability space together with a filtration on it.*

The process (X_n) *is* adapted to the filtration $(\mathcal{F}_n)$ *if for all* $n \in \mathbb{N}$*, the random variable* X_n *is* $\mathcal{F}_n$*-measurable.*

The natural filtration *of a process* (X_n) *is defined by* $\mathcal{F}_0 = \{\emptyset, \Omega\}$ *and* $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ *for* $n \in \mathbb{N}$*.*

Definition 10 (Martingale, supermartingale, submartingale)**.** *Given a filtered probability space* $(\Omega, \mathcal{F}, \mathbb{P})$ *with* $(\mathcal{F}_n)$*, let* $(X_0, X_1, \dots)$ *be a process adapted to the filtration.*

We say (X_n) *is a* martingale *if*

- $\mathbb{E}(|X_n|) < \infty$ *for each* $n \geq 0$*, and*
- $\mathbb{E}(X_n|\mathcal{F}_{n-1}) = X_{n-1}$ *for each* $n \in \mathbb{N}$*.*

The process is a supermartingale *if* $=$ *is replaced by* $\leq$*, and a* submartingale *if* $=$ *is replaced by* $\geq$*.*

If you are in a casino, the game you play at time n may depend on the amount X_{n-1}, or more generally the history $\mathcal{F}_{n-1}$ of your evening, to date. But the casino will ensure that

in expectation you lose: $\mathbb{E}(X_n|\mathcal{F}_{n-1}) \leq X_{n-1}$. This is a supermartingale: it's super for the casino.

At this stage, while it should be clear what the definition 'martingale' means—it is a process which doesn't have any bias; your best guess for the next value is always simply the current value—it shouldn't necessarily be obvious why this is an interesting kind of process to consider. If you pick a real-world problem that you want to model as a random process, you usually will write down some process which obviously isn't a martingale; it maybe tends to grow at first, or so. The short version is: a martingale is the 'random part' of your process, and the 'drift part' can be handled separately. We'll return to this after seeing some examples.

Exercise 7. *Prove that if $Y_1, Y_2, \ldots$ is a sequence of random variables such that for each $n \in \mathbb{N}$ we have $\mathbb{E}(|Y_n|) < \infty$, $\mathbb{E}(Y_n|\mathcal{F}_{n-1}) = 0$, and Y_n is $\mathcal{F}_n$-measurable, then $X_n = \sum_{k=1}^n Y_k$ defines a process adapted to the filtration $(\mathcal{F}_n)$, and this process is a martingale.*

We should remember that the equation $\mathbb{E}(Y_n|\mathcal{F}_{n-1}) = 0$ asserts that the conditional expectation is (almost surely) equal to the constant zero function.

Solution to Exercise 7:

For 'adapted to the filtration', we need to prove that for each $n \geq 0$ the random variable X_n is $\mathcal{F}_n$-measurable. For $n = 0$ this is trivial: X_0 is the constant 0 function which is $\mathcal{F}_0$-measurable (whatever $\mathcal{F}_0$ is). We prove the remaining statements by induction: X_{n-1} is $\mathcal{F}_{n-1}$-measurable, so because $\mathcal{F}_{n-1} \subseteq \mathcal{F}_n$ it is also $\mathcal{F}_n$-measurable, and then $X_n = X_{n-1} + Y_n$ is a sum of $\mathcal{F}_n$-measurable random variables so $\mathcal{F}_n$-measurable.

We have $\mathbb{E}(|X_n|) \leq \sum_{i=1}^n \mathbb{E}(|Y_i|) < \infty$, where the first inequality is the triangle inequality, and the second is because we have a finite sum of real numbers.

For the martingale property, given $n \in \mathbb{N}$ we have

$$\mathbb{E}\big(X_n|\mathcal{F}_{n-1}\big) = \mathbb{E}\big(X_{n-1} + Y_n|\mathcal{F}_{n-1}\big) = X_{n-1} + \mathbb{E}(Y_n|\mathcal{F}_{n-1}) = X_{n-1}$$

where the second equality is linearity of conditional expectation (CE2) to split the conditional expectation of the sum, then (CE1) to evaluate the first part using that X_{n-1} is $\mathcal{F}_{n-1}$-measurable.

Something you need to be careful about is the following: a process being adapted to a filtration has nothing to do with $\mathbb{P}$, but whether that process is a martingale, or a super- or submartingale, does depend on $\mathbb{P}$; if you change to a different measure $\mathbb{Q}$ it will change the conditional expectation.

Another natural way to construct a martingale is the *Doob construction*. Let $(\Omega, \mathcal{F}, \mathbb{P})$ with $(\mathcal{F}_n)$ be a filtered probability space, and X be a random variable with $\mathbb{E}(|X|) < \infty$. The *Doob martingale* is defined by $X_n = \mathbb{E}(X|\mathcal{F}_n)$ for each n.

Exercise 8. *Prove that the Doob construction indeed gives a martingale.*

Solution to Exercise 8:

That $(X_n)_{n\geq 0}$ is adapted to the filtration $(\mathcal{F}_n)_{n\geq 0}$ is immediate from the definition of conditional expectation, property (ii).

Given $n \in \mathbb{N}$ we have

$$\mathbb{E}\big(X_n|\mathcal{F}_{n-1}\big) = \mathbb{E}\big(\mathbb{E}(X|\mathcal{F}_n)|\mathcal{F}_{n-1}\big) = \mathbb{E}(X|\mathcal{F}_{n-1}) = X_{n-1}$$

where the second equality is the tower property (CE9).

We actually need to do a little work here to check the L_1 property.

$$\mathbb{E}(|X_n|) = \mathbb{E}\big(\big|\mathbb{E}(X|\mathcal{F}_n)\big|\big) \leq \mathbb{E}\big(\mathbb{E}(|X|\big|\mathcal{F}_n)\big) = \mathbb{E}(|X|) < \infty$$

as required. There are a couple of steps in here that need explanation, though. The inequality is probably easiest to call an application of the conditional Jensen inequality (CE8), since $x \to |x|$ is a convex function. But you could also write $X = X^+ - X^-$ and use linearity of conditional expectation together with the triangle inequality. And the final inequality looks rather like the tower property, and actually we would usually just say it is the tower property. Formally though (CE9) talks about taking a conditional expectation of a conditional expectation. To deal with the expectation of a conditional expectation, we use a trick: let $\mathcal{G} = \{\emptyset, \Omega\}$, which is a σ-algebra.

Then $\mathbb{E}(|X|\big|\mathcal{G}) = \mathbb{E}(|X|)$ by (CE11): all σ-algebras are independent of $\mathcal{G}$. So we can write

$$\mathbb{E}\big(\mathbb{E}(|X|\big|\mathcal{F}_n)\big) = \mathbb{E}\big(\mathbb{E}(|X|\big|\mathcal{F}_n)\big|\mathcal{G}\big) = \mathbb{E}(|X|\big|\mathcal{G}) = \mathbb{E}(|X|)$$

where the middle equality really is the tower property (CE9) and the outside equalities are the trick we just wrote. Since this works in general, we'll usually simply say that $\mathbb{E}(\mathbb{E}(W|\mathcal{F})) = \mathbb{E}(W)$ is 'the tower property' (CE9).

Example 11. *Let S be a subset of $\{1, 2, \ldots, N\}$ obtained by selecting, independently with probability p for each $1 \leq n \leq N$, the integer n into S. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the obvious probability space of this experiment.*

For each $0 \leq n \leq N$, let $\mathcal{F}_n$ be the σ-algebra generated by $S \cap \{1, 2, \ldots, n\}$ (we can view this as a random variable!), i.e. the random choices for the first n integers.

Let X count the number of 3-term arithmetic progressions in S, and let $X_n = \mathbb{E}(X|\mathcal{F}_n)$ as in the Doob construction. Then $X_0 = \mathbb{E}(X)$, and $X_N = X$, and (X_n) is a martingale sequence.

We'll see the 'point' of this example later: for now it's a rather trivial observation. A Doob martingale is 'trivially' a martingale; what it's doing is saying that we can think of building up to the 'complete' random variable X by steadily taking account of more and more information, revealing a bit of randomness every time.

Example 12. *Consider the* skewed random walk *defined by Y_n taking independently values 1.1 with probability $\frac{1}{2}$ and -0.9 with probability $\frac{1}{2}$. Let $X_n = \sum_{i=1}^n Y_i$ define a process, and let $(\mathcal{F}_n)$ be the natural filtration of the process.*

The process (X_n) is not a martingale, but $(X_n - \frac{1}{10}n)$ is a martingale.

The point here is that we can turn a sum of random variables into a martingale sum: officially, what we did here was subtract the conditional expectation $\mathbb{E}(Y_n|\mathcal{F}_n)$ from Y_n. We might think of the $\frac{1}{10}n$ as the 'drift term' and the martingale sum as the 'random error'

which we expect to be small. In this example, it's very easy to understand the drift term, and there is no randomness involved: all the randomness has ended up in the martingale sum. In more complicated examples, the drift term might have some randomness in it too, but it will in practice be 'better behaved' than the martingale sum.

Example 13. *Let $Y_1, \dots$ be any sequence of random variables with $\mathbb{E}(|Y_n|) < \infty$ adapted to a filtration $(\mathcal{F}_n)$. Then*

$$X_n = \sum_{i=1}^{n} (Y_i - \mathbb{E}(Y_i|\mathcal{F}_{i-1})$$

defines a martingale sequence.

Again, we usually will think of $\sum_{i=1}^{n} \mathbb{E}(Y_i|\mathcal{F}_{i-1})$ as 'the drift term' (even though it is a random variable! it will usually be a lot easier to work with) and the martingale sequence as 'the random error' in the sum $\sum_{i=1}^{n} Y_i$ we want to understand.

We finish off this section with a concentration inequality for martingales.

Theorem 14. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ with $(\mathcal{F}_n)$ be a filtered probability space, and (X_n) a martingale adapted to the filtration. Suppose that for each $n \in \mathbb{N}$ there are numbers a_n, b_n such that $a_n \leq X_n - X_{n-1} \leq b_n$ holds surely.*

Then we have, for any $t > 0$ and $n \in \mathbb{N}$:

$$\mathbb{P}\Big(X_n - X_0 \geq t\Big) \leq \exp\Big(-\frac{2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\Big) \quad \textit{and} \tag{1}$$

$$\mathbb{P}\Big(X_n - X_0 \leq -t\Big) \leq \exp\Big(-\frac{2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\Big). \tag{2}$$

If (X_n) is only known to be a supermartingale, we still obtain (1); if it is only known to be a submartingale then (2) holds.

You should notice that this theorem looks a lot like Hoeffding's inequality: in fact, it's the same thing if the random variables $Y_n = X_n - X_{n-1}$ are independent. But the random increments here don't have to be independent. The proof, however, looks a lot like the proof of Hoeffding's inequality.

Proof. We now prove the first 'upper tail' inequality, and assume in what follows that (X_n) is a supermartingale. The 'lower tail' is the first inequality applied to $(-X_n)$, which is a submartingale.

We first calculate, for some $s > 0$, the moment generating function

$$\mathbb{E}\big(e^{s(X_n - X_0)}\big) = \mathbb{E}\big(e^{s(X_1 - X_0)} e^{s(X_2 - X_1)} \dots e^{s(X_n - X_{n-1})}\big).$$

To do this, we use the tower property (CE9) and 'taking out what is known' (CE10) heavily. We view the expectation above as the result of conditioning first on $\mathcal{F}_{n-1}$, then $\mathcal{F}_{n-2}$, and so on:

$$\begin{aligned}\mathbb{E}\big(e^{s(X_n - X_0)}|\mathcal{F}_{n-1}\big) &= \mathbb{E}\Big(e^{s(X_1 - X_0)} e^{s(X_2 - X_1)} \dots e^{s(X_n - X_{n-1})}|\mathcal{F}_{n-1}\Big) \\ &= e^{s(X_1 - X_0)} \dots e^{s(X_{n-1} - X_{n-2})} \mathbb{E}\Big(e^{s(X_n - X_{n-1})}|\mathcal{F}_{n-1}\Big)\end{aligned}$$

where the second equality is (CE10). At this stage we can use the conditional reverse Jensen inequality (CE8) to write

$$\mathbb{E}\big(e^{s(X_n-X_{n-1})}|\mathcal{F}_{n-1}\big) \le \Lambda_n e^{sa_n} + (1-\Lambda_n)e^{sb_n}$$

where Λ_n satisfies $\mathbb{E}(X_n - X_{n-1}|\mathcal{F}_{n-1}) = \Lambda_n a_n + (1-\Lambda_n)b_n$.

By the same (painful) argument as in Notes 3 Exercise 1 (this is sometimes called Hoeffding's Lemma), we have almost surely

$$\mathbb{E}\big(e^{s(X_n-X_{n-1})}|\mathcal{F}_{n-1}\big) \le e^{s^2(b_n-a_n)/8}\,.$$

We substitute this in:

$$\mathbb{E}\big(e^{s(X_n-X_0)}|\mathcal{F}_{n-1}\big) = e^{s(X_1-X_0)}\dots e^{s(X_{n-1}-X_{n-2})}\cdot e^{s^2(b_n-a_n)^2/8}\,.$$

Now

$$\mathbb{E}\big(e^{s(X_n-X_0)}|\mathcal{F}_{n-1}\big) = \mathbb{E}\Big(\mathbb{E}\big(e^{s(X_n-X_0)}|\mathcal{F}_{n-1}\big)|\mathcal{F}_{n-2}\Big)$$

by (CE9). We can plug in our calculated upper bound for $\mathbb{E}\big(e^{s(X_n-X_0)}|\mathcal{F}_{n-1}\big)$ and obtain by (CE3) an upper bound on this conditional expectation, and repeat the same calculation again. Doing this n times we eventually obtain

$$\mathbb{E}\big(e^{s(X_n-X_0)}\big) = \mathbb{E}\big(e^{s(X_n-X_0)}|\mathcal{F}_0\big) \le \prod_{i=1}^{n} e^{s^2(b_i-a_i)^2/8}\,.$$

This is the same bound as we obtained in the proof of Hoeffding's inequality, and at this stage the proof completes in exactly the same way: we apply Notes 3 Lemma 4, with the same choice of λ, and we get out the same probability bound. Here we used that (by (CE9), repeatedly) we have $\mathbb{E}(X_n) = \mathbb{E}(X_0)$ hence $\mathbb{E}(X_n - X_0) = 0$. □

Theorem 15. *Let X count the number of 3-term arithmetic progressions in a p-random subset S of $\{1,2,\dots,N\}$. Then we have* $\mathbb{P}\Big(|X-\mathbb{E}(X)| \ge t\Big) \le 2\exp\big(-8t^2/9N^3\big)$.

Proof. Let (X_n) be the Doob martingale with filtration $(\mathcal{F}_n)$ as in Example 11. Because $\mathcal{F}_n$ is a finite set for each n, all of whose non-empty sets have positive probability, there is only one version of the conditional expectation $\mathbb{E}\big(X_n|\mathcal{F}_{n-1}\big)$ for each n, calculated as in Exercise 1. That is, once $S \cap \{1,2,\dots,n-1\}$ is revealed, we use the usual formula for conditional expectation, conditioning on this positive probability event.

To bound $|X_n - X_{n-1}|$, we need to ask how great an effect revealing whether n is in S can have on the conditional expectation of X. What it affects is exactly all the 3-term arithmetic progressions that use n: at most, adding or removing n can add or remove all of those progressions. So an upper bound on $|X_n - X_{n-1}|$ is obtained by counting these progressions.

n could appear as the first, second, or third number in such a progression: once we decide on this and the common difference d, which must be in $\{1,2,\dots,N/2\}$, we fixed a progression. So there are at most $3N/2$ progressions that include n, and we conclude

$$-3N/2 \le X_n - X_{n-1} \le 3N/2$$

surely.

Applying Theorem 14 and the union bound (over the positive and negative deviations), we have

$$\mathbb{P}\big(|X-\mathbb{E}(X)|\geq t\big)=\mathbb{P}\big(|X_n-X_0|\geq t\big)\leq 2\exp\Big(-\frac{2t^2}{N(3N/2)^2}\Big)=2\exp\big(-8t^2/9N^3\big)\,. \quad \square$$

We could obtain better bounds with some work. However, it is easy to see that $\mathbb{E}(X)$ will be at least $\frac{1}{20}p^3N^2$ (we didn't try to calculate the constant factor carefully!). We can apply Theorem 15 with $t=\frac{1}{20}\varepsilon p^3N^2$ to obtain a probability

$$2\exp\big(-\varepsilon^2p^6N/450\big)$$

of X being more than a factor $(1+\varepsilon)$, or less than $(1-\varepsilon)$ times, its expectation. Here you should ignore the constant factors: this is an exponential in N bound, which is as good as we could hope for (the probability that $S=\emptyset$ is $(1-p)^{-N}$, which is also exponential in N..!). This is by far the easiest way to prove any such bound on this particular quantity, because there is a lot of dependence: Hoeffding's inequality does not apply.

The sort of bounds we've looked at here are usually called *large deviation* bounds: we're focusing on the probability that some quantity X (that depends on some large parameter N) differs from its expectation by a constant factor (as $N\to\infty$): we want to find that the probability we get will tend to 0 with N rather fast, and we're interested in knowing roughly how fast; here we obtained an exponential convergence, and that was 'good enough'. We didn't really care about the constant factor in the exponent, even though this of course makes a rather big difference to the actual probability!

As previously mentioned, 'classic' probability theory tends to be concerned more with understanding the fine detail: do we have a central limit theorem? what is the distribution 'in the limit'? These kinds of question naturally require us to 'zoom in' and look at the probability that X differs from its expectation by some t which is much smaller than $\mathbb{E}(X)$; typically we'd want to consider t^2 being the same order of magnitude as $\mathrm{Var}(X)$ for these questions.

The 'zoomed in' questions can be quite hard even for relatively simple probabilistic experiments or processes: we need to obtain quite accurate probability estimates. The large deviations behaviour is generally a lot easier to handle (we can happily throw away constant factors or even sub-exponential terms!): in exchange, we'll consider more complicated experiments or processes.

11 Differential equations

In this section we give a general statement that a (well-behaved) process will follow the solution of an associated ordinary differential equation. Here the process will run in discrete time, but the differential equation will describe evolution in continuous time.

It is important to point out that what we are doing here is *not* a 'stochastic differential equation', or even 'stochastic difference equation'. As just mentioned, we're looking at large

deviations: we will try to show that 'most of the probability mass' behaves deterministically when viewed 'from far away'. A stochastic differential (or difference) equation would involve zooming in on the picture until we see a distribution—not just one tiny point that contains most of the probability mass on a big graph, but zoom in to that point until you can see that the probability mass is spread out around its mean—and asking how this distribution evolves over time (if it's a difference equation, over discrete time; if a differential equation, over continuous time). Again, for any given process the second sort of question is 'harder': in return, we'll however be able to handle quite general processes.

To give some intuition about what is going on, think about the usual ordinary differential equation setup. We have a vector $\mathbf{x}$ in $\mathbb{R}^d$, which starts at some $\mathbf{x}(0)$ and evolves over time $t \geq 0$ according to

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = f(t, \mathbf{x})\,.$$

where f is a function from $\mathbb{R}^{d+1} \to \mathbb{R}^d$. To be able to say anything much, we need to impose conditions on f: a standard condition is that f should be *L-Lipschitz-continuous* on some $\mathcal{D} \subseteq \mathbb{R}^{d+1}$, i.e.

$$\big\|f(t, \mathbf{x}) - f(t', \mathbf{x}')\big\|_\infty \leq L\|t - t', \mathbf{x} - \mathbf{x}'\|_\infty$$

where $\|t, \mathbf{x}\|_\infty = \max(|t|, \max_{1\leq i\leq d} |\mathbf{x}_i)|$ is the '∞-norm'. Here $\mathcal{D}$ is some subset of $\mathbb{R}^{d+1}$ (perhaps a large cube). What we should be thinking is that as long as $(0, \mathbf{x}(0))$ is in $\mathcal{D}$, and as long as $(t, \mathbf{x}(t))$ remains in $\mathcal{D}$, we 'understand' the differential equation.

It is a standard fact (which we assume) that under these conditions there exists a unique solution to the differential equation (for as long as the solution remains in $\mathcal{D}$; this can be for all time, or it can be for a bounded time).

Actually, we can say a bit more: if we have two trajectories $\mathbf{x}$ and $\mathbf{z}$ which start at 'close' points in $\mathcal{D}$, then the two trajectories cannot separate 'too fast'. As long as both trajectories are in $\mathcal{D}$, we have

$$\begin{aligned} 0 \leq \|\mathbf{x}(t) - \mathbf{z}(t)\|_\infty &= \Big\|\mathbf{x}(0) - \mathbf{z}(0) + \int_{s=0}^t f(s, \mathbf{x}(s)) - f(s, \mathbf{z}(s))\,\mathrm{d}s\Big\|_\infty \\ &\leq \big\|\mathbf{x}(0) - \mathbf{z}(0)\big\|_\infty + \Big\| \int_{s=0}^t L\|\mathbf{x}(s) - \mathbf{z}(s)\|_\infty\,\mathrm{d}s \end{aligned}$$

and this is now a one-dimensional problem to show that if we have for all $t \geq 0$ the inequalities $0 \leq y(t) \leq C + L\int_{s=0}^t y(s)\,\mathrm{d}s$, then $y(t)$ cannot grow too fast.

Theorem 16 (Grönwall's inequality)**.** *Suppose* $0 \leq y(t) \leq C + L\int_{s=0}^t y(s)\,\mathrm{d}s$ *holds for* $0 \leq t \leq T$*. Then for each* $t \in [0, T]$ *we have* $y(t) \leq Ce^{Lt}$*.*

Proof. Consider the function $m(t) = e^{-Lt}\int_{s=0}^t y(s)\,\mathrm{d}s$, where $t \in [0, T]$. Its derivative is

$$m'(t) = \Big(y(t) - L\int_{s=0}^t y(s)\,\mathrm{d}s\Big)e^{-Lt} \leq Ce^{-Lt}$$

and integrating we get $m(t) \leq \big[\frac{C}{-L}e^{-Ls}\big]_{s=0}^t = \frac{C}{L} - \frac{C}{L}e^{-Lt}$.

By definition we have

$$y(t) \leq C + Le^{Lt}m(t) \leq C + Ce^{Lt} - C = Ce^{Lt}\,. \qquad \square$$

Applying this to our differential equation trajectories, with $C = \|\mathbf{x}(0) - \mathbf{z}(0)\|_\infty$ and L the constant of Lipschitz continuity, we get that provided $\mathbf{x}$ and $\mathbf{z}$ remain in $\mathcal{D}$ on $[0, T]$, then $\|\mathbf{x}(t) - \mathbf{z}(t)\|_\infty \leq Ce^{Lt}$.

This is a rather weak 'stability' type statement: if C is small—the trajectories start close—then they can separate at most exponentially fast. Still, if we can arrange for C to be very tiny, and L is 'reasonable', then Ce^{Lt} can stay quite small for a reasonable length of time.

The relevance of this to processes is as follows. Given $N \in \mathbb{N}$, and a process $(\mathbf{Z})_{n\in\mathbb{N}}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ taking values in $\mathbb{R}^d$, adapted to the filtration $(\mathcal{F}_n)_{n\in\mathbb{N}}$.

Suppose that for each $n \in \mathbb{N}$ we have

$$\mathbb{E}(\mathbf{Z}_n - \mathbf{Z}_{n-1}|\mathcal{F}_{n-1}) = f\big(\tfrac{n-1}{N}, \tfrac{1}{N}\mathbf{Z}_{n-1}\big)\,,$$

where we define the conditional expectation of a random variable taking values in $\mathbb{R}^d$ to be the vector of conditional expectations of the d components.

Then an *ODE associated to the process* is $\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = f(t, \mathbf{x})$, with initial condition $\mathbf{x}(0) = \frac{1}{N}\mathbf{Z}_0$. This does *not* define a unique ODE: we have only specified the value of f at certain points, and in principle we could let f do almost anything between these points. If we do this, we will obtain an f whose Lipschitz continuity is either a very large number or non-existent, and the results to follow will be trivially true. Usually the choice of a 'good' f, with a small Lipschitz continuity constant, will be obvious.

Here we should think of N as being a large number that is used to define $(\mathbf{Z}_n)$.

Example 17. *In a population of size N, at time 0 there are Z_0 infected people. At each time $t \geq 1$, we sample independently (with replacement) people a, b, c in the population. If a is infected at time $t-1$ and b is not, with probability λ person b will be infected at time t. Independently, if c is infected at time $t-1$ then with probability ρ they recover (and become un-infected) at time t.*

Here $d = 1$, hence the normal-face Z.

This model defines a process. In order to avoid any measure-theory issues, let's imagine we stop the process at some large time T; formally, if $t > T$ then istead of the described random evolution we have $Z_t = Z_T$. This lets us write $\Omega = [N]^T$ (we should think of $\omega \in \Omega$ as the vector giving the number of infected people at each time from 1 to T inclusive) and $\mathcal{F} = \mathcal{P}(\Omega)$, and (in principle) we can calculate $\mathbb{P}$. However, (as is normal in this area) we aren't really interested in exactly what probability space supports our process, and in fact 'stopping' the process is unnecessary.

Let $(\mathcal{F}_n)_{n\geq 0}$ be the natural filtration of this process. Observe that for each n, $\mathcal{F}_n$ is a finite collection of events; this means we can define the conditional expectations easily as in Exercise 1:

$$\mathbb{E}\big(Z_n|Z_{n-1} = z\big) = \tfrac{N-z}{N} \cdot \tfrac{z}{N} \cdot \lambda - \frac{z}{N} \cdot \rho = \lambda\frac{Z_{n-1}}{N}\big(1 - \frac{Z_{n-1}}{N}\big) - \rho\frac{Z_{n-1}}{N}\,.$$

So an ODE associated to the process is

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \lambda x(1-x) - \rho x \quad \text{with initial condition } x(0) = Z_0/N.$$

In this example we had $d = 1$; but we might want to consider the spread of multiple diseases, or perhaps model uptake of a vaccine or acquiring of immunity, or even deaths, as well as the spread of infection; then we rapidly need to deal with larger d.

Our hope is that $x(t)N$ is a 'good guess' for the number of infected people at time tN, at least if $t \geq 0$ is not too big.

It's easy to solve this particular ODE: it's a Bernoulli ODE. If $x(0) = 0$ then the solution is $x(t) = 0$ for all $t \geq 0$, while if $0 < x(0) \leq 1$ then we have (as you can check)

$$x(t) = \frac{\lambda - \rho}{\lambda + (\frac{\lambda-\rho}{x(0)} - \lambda)e^{(\rho-\lambda)t}} .$$

(If we had $\lambda = \rho$ there would be a different solution $x(t) = \frac{x(0)}{x(0)t+1}$; we excluded that for simplicity.)

The behaviour of this is very different if $\rho > \lambda$ versus $\rho < \lambda$; in the first case the trajectory rapidly goes to 0, in the latter it rapidly goes to $\frac{\lambda-\rho}{\lambda}$. The latter is a 'dangerous' disease.

It's not hard to check that $f(t, x) = \lambda x(1 - x) - \rho x$ is $(\lambda + \rho)$-Lipschitz continuous on $\mathbb{R} \times [0, 1]$ (this is the maximum absolute value of the derivative).

To see why we might believe that Z_t is likely to be close to $Nx(t)$, observe that in a single time step, we should believe $Z_n - Z_{n-1}$ will likely be very small compared to N; the difference between $f(t, Z_n/N)$ and $f(t, Z_{N-1}/N)$ should be very tiny. The 'difference equation' that the conditional expectation actually satisfies should be very close to an associated ODE. Furthermore, the actual change in one time step will not be exactly the conditional expectation, but it's likely that (at least over several time steps) the randomness 'averages out'. If both these intuitions hold, then we should be able to mimic the weak stability proof for the differential equation to argue that the two quantities cannot separate too fast.

Theorem 18. *Let $\mathcal{D} \subseteq \mathbb{R}^{d+1}$, and let $f(t, \mathbf{x})$ be L-Lipschitz continuous on $\mathcal{D}$ and take values in $\mathbb{R}^d$; suppose that $R \geq 1$ is such that $\|f(t, \mathbf{x})\|_\infty \leq R$ on $\mathcal{D}$. Given a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$, let $(\mathbf{Z}_n)_{n\geq 0}$ be a $\mathbb{R}^d$-valued process on $(\Omega, \mathcal{F}, \mathbb{P})$ adapted to the filtration $(\mathcal{F}_n)_{n\geq 0}$. Suppose*

$$\mathbb{E}\big(\mathbf{Z}_n - \mathbf{Z}_{n-1}|\mathcal{F}_{n-1}\big) = f\big(\tfrac{n-1}{N}, \tfrac{\mathbf{Z}_{n-1}}{N}\big)$$

holds whenever $\frac{1}{N}\mathbf{Z}_{n-1} \in \mathcal{D}$, and suppose $B > 0$ is such that $\big\|\mathbf{Z}_n - \mathbf{Z}_{n-1}\big\|_\infty \leq B$ holds surely.

Now let $\mathbf{x}(t)$ solve the ODE $\mathbf{x}'(t) = f(t, \mathbf{x}(t))$ with initial condition $N\mathbf{x}(0) = \mathbf{Z}_0$. Let $\tau \geq 0$ be such that the following holds: for each $0 \leq t \leq \tau$, the ball of radius $2RN^{-1/4}e^{8Lt}$ around $\mathbf{x}(t)$ is within $\mathcal{D}$.

Then with probability at least $1 - 2d\tau N \exp\big(-\frac{LN^{1/2}eR^2}{2B^2}\big)$, for any $0 \leq n \leq \tau N$, we have

$$\big\|\mathbf{Z}_n - N\mathbf{x}\big(\tfrac{n}{N}\big)\big\|_\infty \leq 2N^{3/4}Re^{8Ln/N} .$$

Before we prove Theorem 18, let's try applying it to the infection model Example 17. We should think of N as being a very large number here, and all the other parameters will turn out to be 'small' constants.

We will take $d = 1$ and $\mathcal{D} = \mathbb{R} \times [0, 1]$; that is, we are interested in any time, but only $x \in [0, 1]$. We'll also look only at the case $\rho < \lambda$ for now, so that the ODE trajectories

converge to $\frac{\lambda-\rho}{\lambda}$, and in fact, let's suppose ρ and λ are such that that number is between 0.1 and 0.9. We'll assume similarly Z_0 is between $0.1N$ and $0.9N$.

Our function f is $f(t,x) = \lambda x(1-x) - \rho x$, as we described earlier. This function actually doesn't depend on t at all: a more complicated model might have time dependence. This f is $(\lambda+\rho)$-Lipschitz continuous on $\mathcal{D}$ (as we checked earlier). We need to find a value for R bounding the maximum absolute value of this function: it's easy to check that the maximum occurs at either $x = 1$ or $x = \frac{\lambda-\rho}{2\lambda}$ (the endpoint $x = 0$ clearly isn't a maximum!) so $R = \max\big(\rho, \frac{1}{4\lambda}(\lambda-\rho)^2\big)$.

Our filtered probability space and process are as discussed in Example 17; we chose f exactly so that the claimed equation for the conditional expectations holds on $\mathcal{D}$.

We can choose $B = 1$: in any time step, at most one person becomes infected and at most one person recovers, for a total change bounded by 1 (If both things happen, the total change is 0..!).

We know a solution to our ODE $x'(t) = f\big(t, x(t)\big)$ with initial value $x(0) = Z_0/N$: it is

$$x(t) = \frac{\lambda-\rho}{\lambda + (\frac{\lambda-\rho}{x(0)} - \lambda)e^{(\rho-\lambda)t}},$$

and (because we decided to look only at this case!) the solution starts and stays within the interval $[0.1, 0.9]$. So the trajectory of the ODE will stay well within $\mathcal{D}$ forever: but what we need to check is how long the ball of a certain radius around the trajectory will stay in $\mathcal{D}$, and this ball grows with t. Because the trajectory stays within $[0.1, 0.9]$, as long as our ball has radius at most 0.1, the ball will be contained in $\mathcal{D}$. Solving

$$0.1 = 2RN^{-1/4}e^{8Lt}$$

we find that we can choose any $\tau \le \frac{1}{8L}\log\big(\frac{1}{20R}N^{1/4}\big)$. It's best to think of this as being more or less a small constant times $\log N$; as N gets large, we'll be able to look at t growing to ∞, albeit rather slowly.

This verifies all the conditions of the theorem: so we get the conclusion. If N is large, the probability bound tends exponentially fast in N to 1, even if we let τ be as large as we possibly can (logarithmic in N); we might as well do that.

The result, then, is the statement that (viewed from far away) our process looks more or less deterministic for a long time: when N is large, and n is not too close to the maximum we are allowed (say, at most $\frac{1}{2}\tau N$) then all but a vanishing fraction of the probability mass is within a vanishingly small distance from the ODE solution. To understand this last comment, let's note that a plot of Z_n against n naturally lives on a graph with y-axis going from 0 to N; the error starts at only $N^{3/4}$ and will only become noticeable on the plot once $2Re^{8Ln/N}$ gets close to $N^{1/4}$: and that only happens once we get to n being something like $N\log N$ (times a small constant). At time say $100N$, that hasn't happened yet: but at the same time, that's a long enough time for our ODE solution to have got very close to the limit $\frac{\lambda-\rho}{\lambda}$ it is converging to: we could reasonably say we've 'seen everything'.

There was obviously nothing 'special' about our choice of $[0.1, 0.9]$. If Nature gave us different values that started in, or converged in, $[0.01, 0.99]$ we would've just slightly changed our estimate for τ.

This is a 'typical' application of Theorem 18. All the constants (d, L, R, B) will usually be 'small numbers' that don't depend on the 'big parameter' N that we think of as growing to ∞; and the function f generally also won't depend on N. Formally none of this is necessary—you can perfectly well apply the theorem in other situations—but this is the normal use case. In this kind of situation, we will always be allowed to take τ to be something on the order of $\log N$, which (because it goes to ∞) means we prove that our process *tracks the ODE* for 'an arbitrarily long time'; whatever feature of the ODE we find interesting, if we take N large enough, we'll see the process exhibits it.

In terms of infection processes, Theorem 18 will always work in more or less the above way: it justifies that our random model more or less behaves deterministically for 'a long time'. If we want to think about the effects of different interventions on a disease, it's enough to solve (perhaps numerically) the associated ODE (which is a great deal easier than the stochastic differential equation: for the former we just have a single point moving, for the latter we have to keep track of the distribution of probability mass over time).

We already described the intuition of the proof. We now give a bit more detail.

There are three sources of 'errors' in the proof, comparing the 'ideal' evolution of $\mathbf{x}$ to the 'actual' evolution of $\mathbf{Z}/N$. First, $\mathbf{x}$ solves a differential equation, while $\mathbf{Z}/N$ is given by a (stochastic) difference equation: the change in a single step is approximately, but not exactly, given by the derivative $\mathbf{x}'/N$. We bound this 'approximation' error in (4) below. Second, in any given time step the actual change will not equal its conditional expectation. We argue that over many time steps these errors average out to a small 'stochastic' error using Azuma's inequality. Third, because there are unavoidable small errors, from very early in the evolution $\mathbf{Z}/N$ is (approximately) following a trajectory of the ODE which does not start at $\mathbf{x}(0)$, but rather a nearby trajectory; the 'stability' error will grow over time.

Although the 'unavoidable' errors are the approximation and stochastic errors—and the stochastic error is much larger—these errors do not grow rapidly over time. Once unavoidable errors have moved the trajectory of $\mathbf{Z}/N$ away from that of $\mathbf{x}$, on the other hand, the stability error does grow, potentially exponentially; it ends up being by far the largest error we have to handle.

We want to deal with this much as with the ODE stability argument. In this case, we will define an *error bound* growing exponentially with n, and argue that, for a given time n, provided that $\mathbf{Z} - N\mathbf{x}$ has not exceeded the error bound at any time before n, then it is unlikely that $\mathbf{Z} - N\mathbf{x}$ exceeds the error bound at time n. We then take a union bound over choices of n. The explicit growing error bound replaces Grönwall's inequality described above.

Proof. Define, for each $y \in \mathbb{R}$, the number $\alpha_y = 2N^{3/4}Re^{8Ly/N}$; this is our error bound at time y.

For each $1 \le i \le \tau N$, define E_i to be the event that $E_1, \dots, E_{i-1}$ do not occur, but $\left|\mathbf{Z}_i - N\mathbf{x}(i/N)\right\|_\infty > \alpha_y$. Observe E_i is $\mathcal{F}_i$-measurable for each $i \in \mathbb{N}$.

We claim that for each $1 \le i \le \tau N$, if no event $E_1 \cup \dots \cup E_{i-1}$ occurs, then we have

$$\left|\mathbb{E}(\mathbf{Z}_i - \mathbf{Z}_{i-1}|\mathcal{F}_{i-1}) - f\left(\tfrac{i-1}{N}, \mathbf{x}(\tfrac{i-1}{N})\right)\right| \le \tfrac{2L}{N}\alpha_{i-1}\,. \tag{3}$$

To see this, observe that since none of the events $E_1, \dots, E_{i-1}$ occurs, in particular $\|\mathbf{Z}_{i-1} - N\mathbf{x}((i-1)/N)\|_\infty \le \alpha_{i-1}$, so $\frac{\mathbf{Z}_{i-1}}{N}$ is in $\mathcal{D}$. By L-Lipschitz continuity of f, we have

$$\big\| f\big(\tfrac{i-1}{N}, \tfrac{\mathbf{Z}_{i-1}}{N}\big) - f\big(\tfrac{i-1}{N}, \mathbf{x}(\tfrac{i-1}{N})\big)\big\|_\infty \le \tfrac{L}{N}\alpha_{i-1}\,.$$

Putting this together with our assumption

$$\mathbb{E}\big(\mathbf{Z}_i - \mathbf{Z}_{i-1} | \mathcal{F}_{i-1}\big) = f\big(\tfrac{i-1}{N}, \tfrac{\mathbf{Z}_{i-1}}{N}\big)$$

we get (3).

Our next claim is that for each $1 \le i \le \tau N$ we have

$$\Big\| N\mathbf{x}\big(\tfrac{i}{N}\big) - N\mathbf{x}\big(\tfrac{i-1}{N}\big) - f\big(\tfrac{i-1}{N}, \mathbf{x}(\tfrac{i-1}{N})\big)\Big\|_\infty \le \tfrac{LR}{N}\,. \tag{4}$$

Consider the jth coordinate of $\mathbf{x}(t)$, which is a differentiable function of t on $[\frac{i-1}{N}, \frac{i}{N}]$. The Mean Value Theorem says there is c in this interval such that

$$N\mathbf{x}\big(\tfrac{i}{N}\big) - N\mathbf{x}\big(\tfrac{i-1}{N}\big) = \mathbf{x}'(c)_j = f(c, \mathbf{x}(c))_j\,.$$

In each coordinate, $\mathbf{x}(c)$ differs from $\mathbf{x}\big(\frac{i-1}{N}\big)$ by at most $\frac{R}{N}$ and is therefore in $\mathcal{D}$, so by L-Lipschitz continuity we have $\big|f(c, \mathbf{x}(c))_j - f\big(\frac{i-1}{N}, \mathbf{x}\big(\frac{i-1}{N}\big)\big)_j\big| \le \frac{LR}{N}$ as in (4). We should really check this 'is in $\mathcal{D}$' claim. What we want to argue is $\frac{R}{N} \le 2N^{-1/4}Re^{8L(i-1)/N}$. This is true since $e^{8L(i-1)/N} \ge 1$, and $2N^{-1/4}R \ge RN^{-1}$ holds for all $N \ge 1$.

Having established these two claims, we now want, for a fixed $1 \le n \le \tau N$, to estimate $\mathbb{P}(E_n)$. For each $1 \le i \le n$, let $\mathbf{Y}_i = \mathbf{Z}_i - \mathbf{Z}_{i-1} - \mathbb{E}(\mathbf{Z}_i - \mathbf{Z}_{i-1}|\mathcal{F}_{i-1})$. Since $\|\mathbf{Z}_i - \mathbf{Z}_{i-1}\|_\infty \le B$, the same bound holds on the conditional expectation and so $\|\mathbf{Y}_i\|_\infty \le 2B$ holds surely.

For each $1 \le j \le d$, the sequence $\sum_{i=1}^m (\mathbf{Y}_i)_j$ (i.e. the sums of the jth components) for $0 \le m \le n$ is a martingale sequence, by Exercise 7. Azuma's inequality, Theorem 14, therefore applies: we obtain

$$\mathbb{P}\Big(\Big|\sum_{i=1}^n (\mathbf{Y}_i)_j\Big| \ge \tfrac{1}{4}\alpha_n\Big) \le 2\exp\Big(-\frac{\alpha_n^2}{128nB^2}\Big)\,,$$

where we recall that the range of each $(\mathbf{Y}_i)_j$ is $[-2B, 2B]$. By the union bound over the d choices of j, we obtain

$$\mathbb{P}\Big(\Big\|\sum_{i=1}^n \mathbf{Y}_i\Big\|_\infty \ge \tfrac{1}{4}\alpha_n\Big) \le 2d\exp\Big(-\frac{\alpha_n^2}{128nB^2}\Big)\,.$$

We claim E_n is contained in the set of $\omega \in \Omega$ such that $\big\|\sum_{i=1}^n \mathbf{Y}_i\big\|_\infty > \frac{1}{4}\alpha_n$, in which case the equation above gives

$$\mathbb{P}(E_n) \le 2d\exp\Big(-\frac{\alpha_n^2}{128nB^2}\Big) \tag{5}$$

Indeed, suppose $\omega \in \Omega$ is such that $\Big\| \sum_{i=1}^n \mathbf{Y}_i \Big\|_\infty \le \frac{1}{4}\alpha_n$ holds, and we aim to show $\omega \notin E_n$. We have

$$\Big\| \sum_{i=1}^n \mathbf{Y}_i \Big\|_\infty = \Big\| \sum_{i=1}^n \mathbf{Z}_i - \mathbf{Z}_{i-1} - \mathbb{E}(\mathbf{Z}_i - \mathbf{Z}_{i-1}|\mathcal{F}_{i-1}) \Big\|_\infty \in \big[-\tfrac{1}{4}\alpha_n, \tfrac{1}{4}\alpha_n \big]$$

$$\text{so} \quad \Big\| \mathbf{Z}_n - \mathbf{Z}_0 - \sum_{i=1}^n f\big(\tfrac{i-1}{N}, \mathbf{x}(\tfrac{i-1}{N})\big) \Big\|_\infty \in \big[-\tfrac{1}{2}\alpha_n, \tfrac{1}{2}\alpha_n \big]$$

$$\text{so} \quad \Big\| \mathbf{Z}_n - \mathbf{Z}_0 - (N\mathbf{x}(0) - N\mathbf{x}(n/N)) \Big\|_\infty = \Big\| \mathbf{Z}_n - N\mathbf{x}(n/N) \Big\|_\infty \in [-\alpha_n, \alpha_n]$$

Here the first line is our assumption on $\sum_i \mathbf{Y}_i$ together with writing out the definition. For the second line, observe that if $\omega \in E_1 \cup \cdots \cup E_{n-1}$ then $\omega \notin E_n$ by definition and we are done: so suppose $\omega \notin E_1 \cup \cdots \cup E_{n-1}$. Then we can apply (3) and the triangle inequality to obtain the second line. Here we use that since α_y is non-negative and increasing in y,

$$\sum_{i=1}^n \tfrac{2L}{N}\alpha_{i-1} \le \int_{y=-\infty}^n \tfrac{2L}{N}\alpha_y \, \mathrm{d}y = \tfrac{2L}{N} \cdot \tfrac{N}{8L}\alpha_n = \tfrac{1}{4}\alpha_n \,.$$

For the final line, we similarly use (4) and the triangle inequality, where the sum of the n 'errors' is $n\frac{LR}{N} \le \frac{1}{2}\alpha_n$. Now the final line exactly says $\omega \notin E_n$, so (5) holds. Again, we should explain why the claimed inequality $\frac{nLR}{N} \le \frac{1}{2}\alpha_n$ is true. Substituting in our value for α_n, we want to prove

$$\tfrac{nLR}{N} \le N^{3/4} R e^{8Ln/N} \,.$$

The easiest way to see this is to consider it as a statement about continuous $n \in \mathbb{R}_{\ge 0}$ and use calculus. The inequality clearly holds for $n = 0$ (the LHS is zero and the RHS is $N^{3/4}R > 0$) and the derivative of the LHS with respect to n is LR/N, while of the RHS is $8LN^{-1/4}Re^{8Ln/N} \ge 8LN^{-1/4}R$ which is larger, so by the Mean Value Theorem the inequality holds for all $n \ge 0$.

Finally, if $\omega \notin E_1 \cup \cdots \cup E_{\tau N}$ then by definition we have $\big\| \mathbf{Z}_n - N\mathbf{x}\big(\frac{n}{N}\big) \big\|_\infty \le \alpha_n = 2N^{3/4}Re^{4Ln/N}$ for all $1 \le n \le \tau N$, as required by the theorem. The claimed probability bound is simply the union bound:

$$\mathbb{P}\big(E_1 \cup \cdots \cup E_{\tau N}\big) \le \sum_{n=1}^{\tau N} 2d \exp\Big(-\frac{N^{3/2}R^2 e^{16Ln/N}}{32nB^2} \Big) \le 2d\tau N \exp\big(-\tfrac{LN^{1/2}eR^2}{2B^2} \big) \,.$$

For the final inequality here, we used the fact that $e^{16Ln/N}/n$ is minimised at $n = N/16L$. $\square$

Exercise 9. *Use Theorem 18 to analyse what happens in Example 17 if $\rho > \lambda$. In order to obtain τ similar to the $\lambda < \rho$ case, you will need to change $\mathcal{D}$: can you do so without changing L and R?*

Solution to Exercise 9:

There are actually several ways to answer this. Maybe the simplest is to observe that neither the process nor the ODE solution can leave $\mathcal{D}$, so that in fact all the points we want to claim are in $\mathcal{D}$, are really in there without changing anything. But of course this requires

us to look into the proof of Theorem 18 and check it works in this particular case (this is *not* something that would work for all processes!).

We could also look carefully at the ODE solution and make τ smaller so that we stop when the ball radius touches the boundary of $\mathcal{D}$. This works fine, but we 'lose' in that τ will be smaller.

We could make $\mathcal{D}$ a bit larger, for example $\mathbb{R} \times [-0.1, 1.1]$. If we *only* do this, then we will have to change L and R, though, for the worse. But here we can play a trick: we let f be defined as it is on $\mathbb{R} \times [0, 1]$, but set $f(t, x) = f(t, 0)$ for all $x < 0$, and $f(t, x) = f(t, 1)$ for all $x > 1$. This avoids needing to change L and R, but because neither the process nor the ODE solution leaves the original domain, this change to f doesn't change the ODE trajectories or the process. This gets the same result as the first route, but we don't have to look in to the proof of Theorem 18: we're simply applying the statement.

In Hoeffding's and Azuma's inequalities, there is a *trade-off*: we can choose the amount of deviation from the expectation we will tolerate. If we want to keep the deviation small, we get a weaker probability bound; if we allow bigger deviations, we get a better probability bound. In Theorem 18 we don't have such a trade-off, but we can ask for it.

Exercise 10. *Prove that we can modify Theorem 18 as follows: Fix any* $\frac{1}{2} < \nu < 1$. *We choose* τ *such that the ball of radius* $2RN^{\nu-1}e^{8Lt}$ *around* $x(t)$ *is in* $\mathcal{D}$ *whenever* $0 \le t \le \tau$, *and we obtain a probability bound* $1 - 2d\tau N \exp\big(-\frac{LN^{2\nu-1}eR^2}{2B^2}\big)$ *for the event that for all* $0 \le n \le \tau N$ *we have*

$$\big\|\mathbf{Z}_n - N\mathbf{x}\big(\tfrac{n}{N}\big)\big\|_\infty \le 2N^\nu Re^{8Ln/N}\,.$$

Hint: You should find that this does not require you to change the proof much: it is enough to explain where changes are needed, and what calculations need to change as a result. You may need to assume N *is sufficiently large, though.*

Solution to Exercise 10:

We modify as follows. First, we redefine $\alpha_y = 2N^\nu Re^{8Ly/N}$; that is, we replace the exponent $\frac{3}{4}$ with ν. In fact, that's what we've done for the entire statement: we replaced $\frac{3}{4}$ with ν in exponents.

The hope is that we can now more or less just go through the proof without needing to change much more. Indeed, our two claims that give (3) and (4) don't need any changing: the only two places where the proof as written interacts with the changes we made at all is that we need to check certain points are in $\mathcal{D}$. In the proof of (3), the justification is that our 'second point' is within α_{i-1} of the ODE solution: but since we also changed in the statement the size of the ball about the ODE solution that is required to stay in $\mathcal{D}$ in a matching way, this remains true. In the proof of (4) we similarly need that our ball has radius at least $\frac{R}{N}$; this would be true even if we were allowed to take $\nu = 0$, and it certainly is for $\nu > \frac{1}{2}$.

For (5), while α_n appears all over the place in the derivation of this inequality, we don't actually ever substitute in its value, so the inequality will still hold for the changed definition.

This leaves us with the three lines of estimates and the final probability bound to check.

The first line still just follows from our assumption on the martingale sum. For the second line, we need to check the calculation upper bounding the sum by an integral and

doing the integral still works. Which it does: the $N^{3/4}$ which has been replaced by N^{μ} is just a constant factor that appears multiplying all four terms in the calculation, so nothing really changed. The final line needs us to check $\frac{nLR}{N} \leq \frac{1}{2}\alpha_n$: here we really should check this works. Actually, we really should've checked it in the original proof too..! We need

$$\frac{nLR}{N} \leq N^{\nu} R e^{8Ln/N} .$$

The R cancels on both sides, and substituting $y = Ln$ we need to check

$$y \leq N^{1+\nu} e^{8y/N} .$$

The derivative of the RHS term with respect to y is $8N^{\nu}e^{8y/N} \geq 8$ (if $y \geq 0$), which is larger than the LHS derivative, so it's enough by the Mean Value Theorem to check the inequality holds for $y = 0$, which it does. Again, we just needed $\nu \geq 0$ for this.

The final paragraph doesn't really need any modification: the new value of α_n turns out to be exactly what the theorem requires for the error between the random process and the ODE solution (by design!) and the union bound calculation works pretty much verbatim, we just replace $N^{3/2}$ by $N^{2\nu}$ when we substitute in α_n^2 and this gives $N^{2\nu-1}$ in the final RHS probability bound, as we required.

It's actually not clear why we insisted on $\frac{1}{2} < \nu < 1$ in this question at this stage: we seem never to have needed these bounds, just $\nu \geq 0$..!

For the lower bound, the reason is that if $\nu \leq \frac{1}{2}$, the probability bound we get is smaller than 0 even if $\tau N = 1$. Saying 'the probability is at least a negative number' is vacuously true. The statement is still correct, it just doesn't say anything useful.

For the upper bound, actually we can and it can even (rarely!) be useful to allow $\nu \geq 1$. However, in this case, if we're thinking of $\mathbf{Z}$ as taking values on the order of N, even at $n = 1$ the error estimate we get is larger than $2RN$ and it increases fast: this usually means the 'error term' is much bigger than the process we're trying to control, and the result is usually no use.

In our example, we had a 'nice' process which conveniently had an associated ODE which was very easy to solve. In general, we can't hope for that (but we may still be able to solve the ODE numerically or analytically prove some properties of it). However, something which could have happened (our example deliberately avoided it!) is that our values for $f(t, \mathbf{x})$ have some very small dependence on N, or they are very close to being 'nice' functions but aren't quite. This happens due to things like rounding errors: N and $N-1$ are 'almost' the same, but we might still have a fraction $\frac{N}{N-1}$ appear in a conditional probability, or we might want to choose some parameter of the model equal to $\frac{1}{2}N$, but if N is odd, because this isn't an integer we have to round it.

Exercise 11. *We can modify Theorem 18 slightly as follows: there is $\delta > 0$ (which you should determine, and which will depend on N) such that we insist only on*

$$\left\| \mathbb{E}\big(\mathbf{Z}_n - \mathbf{Z}_{n-1} | \mathcal{F}_{n-1}\big) - f\big(\tfrac{n-1}{N}, \tfrac{\mathbf{Z}_{n-1}}{N}\big) \right\|_{\infty} < \delta$$

holding whenever $\frac{1}{N}\mathbf{Z}_{n-1} \in \mathcal{D}$. Furthermore we do not insist on $\mathbf{x}(0) = \frac{1}{N}\mathbf{Z}_0$, but only $\left\|\mathbf{x}(0) - \frac{1}{N}\mathbf{Z}_0\right\|_{\infty} < \delta$. We say that the ODE $\mathbf{x}'(t) = f(t, \mathbf{x}(t))$ with initial condition $\mathbf{x}(0)$

is δ-associated to the process $(Z_n)_{n\geq 0}$. *Prove that we still obtain the same conclusion in Theorem 18 if we only ask for a* δ*-associated ODE, provided* $\delta > 0$ *is sufficiently small.*

Hint: Again, you should not need to change the proof much. In fact, the calculations already take account of the extra errors: why?

Solution to Exercise 11:

As with the previous exercise, you should go through the entire proof and see what needs to be done.

It turns out there are three places where things change.

We need to check that $\left|\mathbf{Z}_0 - N\mathbf{x}(0)\right| \leq \alpha_0$ is true, because we use this assumption (it's not covered by the inductive proof of the rest!). This means we need $\delta N \leq \alpha_0$.

The derivation of (3) needs to change, because we replace the equality $\mathbb{E}(\mathbf{Z}_i - \mathbf{Z}_{i-1}|\mathcal{F}_{i-1}) = f\left(\frac{i-1}{N}, \frac{\mathbf{Z}_{i-1}}{N}\right)$ by

$$\left|\mathbb{E}(\mathbf{Z}_i - \mathbf{Z}_{i-1}|\mathcal{F}_{i-1}) - f\left(\tfrac{i-1}{N}, \tfrac{\mathbf{Z}_{i-1}}{N}\right)\right| < \delta\,.$$

But we can use the 'slack' built in to the proof: the factor 2 in (3) wasn't necessary, and we get an error bounded by $\frac{L}{N}\alpha_{i-1} + \delta \leq \frac{2L}{N}\alpha_i$ here; we need $\delta < \frac{L}{N}\alpha_0$. for this to be a true statement.

Finally, the last line of the approximation

$$\left\|\mathbf{Z}_n - \mathbf{Z}_0 - (N\mathbf{x}(0) - N\mathbf{x}(n/N))\right\|_\infty = \left\|\mathbf{Z}_n - N\mathbf{x}(n/N)\right\|_\infty \in [-\alpha_n, \alpha_n]$$

is no longer valid, because $\mathbf{Z}_0 = N\mathbf{x}(0)$ isn't exactly true any more: there will be a further error δN here.

To handle this, we need to slightly improve either some previous error bound, or to fit it in to the final error. We'll do the latter. We need $\frac{nLR}{N} + \delta N \leq N^{3/4}Re^{8Ln/N}$ to hold for all n. The same calculus argument as in the proof of Theorem 18 says we just need to check this for $n = 0$, where it is true provided $\delta N \leq N^{3/4}R$, i.e. if $\delta \leq RN^{-1/4}$.

Putting these three requirements on δ, we end up with $\delta = \min(1, L)RN^{-1/4}$.

The result of this exercise allows us to ignore 'annoying' ODEs as described above: we can afford to write the 'nice' ODE that ignores rounding errors and such, and it will still be a δ-associated ODE of the process, for some δ which will (in practice) be small enough for Exercise 11. If we want, we can even combine Exercises 10 and 11 and ask for both strengthenings.

What happens if we want to look at a process which runs for more than τN steps? We cannot use Theorem 18 to answer the question directly, and the reason is that (even if $\mathcal{D}$ is big enough to contain our long trajectories) if we are dealing with an associated ODE which is not stable—its trajectories really do separate exponentially fast—then (while we weren't all that efficient in calculation) if we try to look well after time τN, the probability mass genuinely will be spread out over a 'large' region of $\mathcal{D}$.

What if this isn't the case? Consider Example 17 with $\lambda > \rho$. The associated ODE is in fact stable: its trajectories from any positive starting point converge exponentially fast, so the heuristic above doesn't apply. Can we do better? In this case, because there is no time dependence in the ODE, there is a convenient trick to improve a lot.

Exercise 12. *Take the process of Example 17. Fix $\frac{1}{2} < \nu < 1$. Consider applying Exercise 10 repeatedly, with a suitable choice of τ, as follows: analyse the process up to time τN, then let $Z_{\tau N} = Z'_0$ be the start point for a (randomly starting) new process evolving according to the same rule which we again analyse up to time τN, and so on. Explain why this process is still simply $(Z_n)_{n\geq 0}$.*

Using this, argue that the following is true for some number T^*. Suppose $\lambda = \frac{1}{2}$ and $\rho = \frac{1}{4}$, and $Z_0 = \frac{3}{4}N$. With probability at least 0.99, for all $0 \leq n \leq T^*$ we have $0.1N \leq Z_n \leq 0.9N$. How large can you make T^*? What value should ν take to optimise it?

Hint: You should use the rapid convergence of $x(t)$ to $\frac{1}{2}$. This is sometimes called 'catch and throw' analysis: when you see the errors begin to grow so that you are concerned the process is going off track, 'catch' it. Then restart from the position you observe: 'throw' it back in the correct direction (the ODE converges rapidly to $\frac{1}{2}$) and repeat.

Solution to Exercise 12:

The process we defined which 'stops and restarts' is still the same process as $(Z_n)_{n\geq 0}$ because the update rule does not look at the time or the history of the process at earlier states. The 'restarted' process from a given distribution $Z'_0 = Z_{\tau N}$ gives a distribution for Z'_1 which is identical to the distribution for $Z_{\tau N+1}$, and so on.

For the 'catch-and-throw' analysis, let's write down the ODE solution in our specific case. For any given starting point y_0, the solution of $x' = \frac{1}{2}x(1-x) - \frac{1}{4}x$ starting at y_0 is

$$x(t) = \frac{1}{2 + (1/y_0 - 2)e^{-t/4}}\,.$$

Our initial choice is $y_0 = \frac{3}{4}$, and we can notice that at time $t = 8$ (this is just chosen to cancel the division by 4 in the exponential and then some!) the solution gets relatively close to $\frac{1}{2}$. In fact, the 'worst' starting points in $[0.1, 0.9]$ are going to be the endpoints (the trajectories don't cross!) which go to respectively 0.324.. and 0.531...

So what we notice is this: wherever in $[0.2, 0.8]$ we start, if we at time $t = 8$ are within a ball of radius 0.1 about the ODE solution, then in particular we are still in $[0.2, 0.8]$ (and we can repeat the argument). This looks like the kind of statement Theorem 18 will say is a likely outcome. In order to get a better probability bound, let's use Exercise 10. Because we are looking at a rather small time (we are taking $\tau = 8$) and we are looking at a rather big ball (the radius is 0.1, not anything tending to zero with N) we can afford to let ν be close to 1; this will give us the best probability bound. We'll figure out later what exactly we can take ν to be.

When we apply Exercise 10, we do so with the same $d = 1$, $\mathcal{D} = \mathbb{R} \times [0, 1]$, $L = \lambda + \rho = \frac{3}{4}$, $R = \frac{1}{4}$ and $B = 1$ as we used for the application of Theorem 18 for this process. However we'll choose $\tau = 8$; and our choice of ν will ensure that at $n = 8N$ our error bound $2N^\nu Re^{8Ln/N}$ is at most $0.1N$. (We still need to figure out what ν makes that true!) Because our ODE trajectory starts within, and (as we argued above) stays within $[0.2, 0.8]$, this means the 'ball remains in $\mathcal{D}$' condition will be satisfied.

So we can quote the outcome: Letting $(Z'_n)_{n\geq 0}$ be the process starting at any given $y_0 N$ with $0.2 \leq y_0 \leq 0.8$, the following holds. With probability at least $1 - 16N \exp\big(-\frac{3N^{2\nu-1}e}{128}\big)$, for all $0 \leq n \leq 8N$ we have

$$\big|Z'_n - Nx\big(\tfrac{n}{N}\big)\big| < 0.1$$

and in particular, if the likely event occurs, Z'_{4N} is within 0.1 of $Nx(4)$, which (as we calculated above) is in $[0.324.., 0.531..]$. So $0.2N \le Z'_{4N} \le 0.8N$; and (because the ODE solution doesn't leave $[0.2, 0.8]$) Z'_n remains in $0.1N, 0.9N]$ for all $0 \le n \le 8N$. For convenience, let's write $p_\nu = 16N \exp\big(-\frac{3N^{2\nu-1}e}{128}\big)$, and immediately observe that for large N we have $p_\nu \ge \exp\big(-N^{2\nu-1}/500\big)$. (This is easy to check; and holds for 'realistic' N; this is just done to make things simpler later, even if it 'loses' us a bit).

This is the critical point of the catch-and-throw analysis. We now need to explain why we can 'catch and throw'. Suppose Z'_0 is not started from a specific fixed y_0N with $y_0 \in [0.2, 0.8]$, but instead from a distribution of numbers about which we know nothing except that the support is contained in $[0.2N, 0.8N]$; this is the distribution we get after running to time $8N$ and 'catching' it. We will 'lose' the small amount of probability mass where the random process is not in this interval after $8N$ steps.

We claim that the probability, from this random starting distribution Z'_0, of getting Z'_{8N} outside $[0.2N, 0.8N]$, is at most the same p_ν we saw above. The reason is: suppose we sample from the distribution Z'_0. We get a start y_0, and (whatever sample we got) we have

$$\mathbb{P}\big(Z'_{8N} \notin [0.2N, 0.8N] \big| Z'_0 = y_0\big) \le p_\nu\,,$$

by Exercise 10. But the events $Z'_0 = y$ are disjoint, so we can write

$$\begin{aligned}
&\mathbb{P}\big(Z'_{8N} \notin [0.2N, 0.8N] \big| Z'_0 \in [0.2N, 0.8N]\big)\\
&= \sum_{y_0=0.2N}^{0.8N} \mathbb{P}\big(Z'_{8N} \notin [0, 2N, 0.8N] \big| Z'_0 = y_0\big) \cdot \mathbb{P}(Z'_0 = y_0 \big| Z'_0 \in [0.2N, 0.8N])\\
&\le p_\nu \sum_{y_0=0.2N}^{0.8N} \mathbb{P}(Z'_0 = y_0 \big| Z'_0 \in [0.2N, 0.8N])\\
&= p_\nu \mathbb{P}(Z'_0 \in [0.2N, 0.8N] \big| Z'_0 \in [0.2N, 0.8N]) = p_\nu\,.
\end{aligned}$$

Now we are in a position to use this inductively. We'll apply it with Z'_0 being Z_0, then Z_{8N}, then Z_{16N}, and so on: we get

$$\mathbb{P}\big(Z_{8kN} \in [0.2N, 0.8N] \big| \big| Z_{8(k-1)N} \in [0.2N, 0.8N]\big) \ge 1 - p_\nu$$

for each $k \in \mathbb{N}$.

Multiplying the conditional probabilities, we get

$$\mathbb{P}\big(Z_{8kN} \in [0.2N, 0.8N]\big) \ge \big(1 - p_\nu\big)^k\,,$$

and we just need to see how big k can be before this goes below 0.99. We have $1 - p_\nu \ge e^{-2p_\nu}$ by considering a Taylor expansion (and that p_ν is very small), so we want to solve

$$e^{-2kp\nu} = 0.99 \quad \text{so} \quad 2kp_\nu = 0.01$$

and we get that $k = 0.005 \exp\big(N^{2\nu-1}/500\big)$ works.

Let's now see how big we could afford to take ν. We needed

$$2N^\nu R e^{64L} < 0.1N \quad \text{so} \quad N^{1-\nu} > 5e^{48}$$

and so $(1-\nu)\log N = 50$ will do; we can have $\nu = 1 - \frac{50}{\log N}$. Plugging that in, we get

$$k = 0.005 \exp\left(N^{1-100/\log N}/500\right) = 0.005 \exp\left(Ne^{-100}/500\right),$$

and finally what this tells us is that we can take

$$T^* = 8kN = 0.04N \exp\left(Ne^{-100}/500\right).$$

This is a fairly horrible formula: but the 'point' is that it is something growing exponentially in N. We could have been a good deal more careful and improved the constants, but there is no hope to do better than an exponential in N function.

Formally the above probability equations only talk about the value of Z at multiples of $8N$, but the probability bound is actually (as we said above) looking at the event that the process stays within $[0.1N, 0.9N]$ for the entire time and finishes within $[0.2N, 0.8N]$, so what we end up with is that for this choice of k we have

$$\mathbb{P}\big(Z_n \in [0.1N, 0.9N] \text{ for all } 0 \le n \le T^*\big) \ge 0.99\,.$$

This is what we wanted.

Let's briefly point out why there is 'no hope' of getting a better bound than exponential. We've argued that it is really likely—probability better than 0.99—that our random process stays in $[0.1N, 0.9N]$ for an exponentially long time. But of course, it is possible that at some point we see a long sequence of times when no-one gets infected but people do recover. While there are more than $0.05N$ infected people, the chance of a recovery at any given time is at least 0.05 suppose this happens. If in addition the λ-chance of infection doesn't occur, and these both occur for $0.85N$ time steps one after another, we get down to at most $0.05N$ infected people. The chance of that happening is $(0.05(1-\lambda))^{0.85N}$, which is also exponential in N.

Consider splitting time up into disjoint consecutive intervals of length $0.85N$, and consider the event that after k intervals, at the end of each interval the process is in $[0.1N, 0.9N]$. The probability, conditioned on this occurring, that it remains in $[0.1N, 0.9N]$ after the $(k+1)$st interval is at most $(0.05(1-\lambda))^{0.85N}$. Multiplying the conditional probabilities, the probability that the process remains in $[0.1N, 0.9N]$ up to time $k \cdot 0.85N$ is at most

$$\big(1 - (0.05(1-\lambda))^{0.85N}\big)^k \le \exp\big(-\tfrac{k}{2}(0.05(1-\lambda))^{0.85N}\big),$$

and if $k = 2(0.05(1-\lambda))^{-0.85N}$, this becomes at most $e^{-1} < 0.99$.

In general, there are three (broadly) different ways an ODE can behave—in each case, really we mean 'behaviour on a domain $\mathcal{D}$'.

One is 'unstable', or 'chaotic'. The simple one-dimensional example is $f(t,x) = x$. This is 1-Lipschitz continuous; the trajectories are $x(t) = x_0 e^t$. The trajectories genuinely do separate exponentially fast; just as fast as the worst-case estimate we proved with Grönwall's inequality; they are pushed away from each other 'as fast as possible'. A process whose associated ODE behaves like this can still be tracked—Theorem 18 still applies—but the probability mass really will spread out exponentially fast. It won't really be quite as fast-growing as the $2N^{3/4}Re^{8Ln/N}$ bound we gave—we were a bit lazy in the analysis—but we

would only be able to improve some of the constants a bit in general. This kind of exponential growth corresponds to needing to take τ on the order of $\log N$; we can track the process for something like $N \log N$ steps.

Another regime is 'neutral'. This sort of regime doesn't really show up much in practice (if you want to apply Theorem 18; if you are doing SDEs then it certainly does!) the one-dimensional version is $f(t,x)$ is close to being a constant (in higher dimensions, things are more complicated! Basically, trajectories don't tend to either spread apart much or come together.). The obvious 'good example' is $f(t,x) = 0$, which describes a drift-free random walk (not necessarily the simple walk that goes up 1 or down 1 with equal probability, summing any mean-zero distribution will do). Here (usually!) one can improve Theorem 18 quite a lot; we would expect that 'really' the probability mass spreads out roughly with n^α for some $\alpha > 0$ (for the simple random walk, we'd have $\alpha = 0.5$) and correspondingly we'd expect to be able to take $\tau = N^{1-\alpha}$ and therefore track the process for something like $N^{2-\alpha}$ steps. In one dimension, this regime corresponds to L tending to zero with N, and Theorem 18 will actually give quite acceptable results (not as good as 'we expect', but not too bad either). In higher dimensions, one can have neutral ODEs without L being small (for example trajectories that go in circles) and analysing these can get quite painful.

Exercise 13. *Use Theorem 18 to analyse the simple random walk; that is, $Z_0 = 0$ and for each n the increment $Z_n - Z_{n-1}$ is equal to either 1 or -1, with equal probability, independently.*

Hint: if you set $L = 0$ then Theorem 18 gives only trivial results. Why? and how small can you set L to be?

Solution to Exercise 13:

If we put $L = 0$ in Theorem 18, the probability bound we get is $1 - 2\tau N$, which will be negative already for $\tau N = 1$: so we don't get any guarantee of being able to analyse even one step of the process. The same goes for trying to take $R = 0$.

The associated ODE of this process is $\frac{\mathrm{d}x}{\mathrm{d}t} = 0$; we can take $d = 1$, $\mathcal{D} = \mathbb{R}^2$, $B = 1$ and $R = 1$. (It's not really obvious this is an optimal choice, but let's go with it.)

In order for the probability bound to be non-trivial, we need the exponent to tend to 0 with N, fast enough to beat the τN term. Let's therefore set $L = N^{\gamma - 1/2}$, where $\gamma > 0$. Our probability bound is $1 - 2\tau N \exp\big(-\frac{N^\gamma e}{2}\big)$, which tends to 1 as $N \to \infty$ if $\gamma > 0$ is fixed.

The ball condition is vacuous since $\mathcal{D} = \mathbb{R}^2$, so we really only care that our error bound stays reasonably small; say less than N. In other words, we want

$$2n^{3/4} e^{8nN^{\gamma - 3/2}} < N\,.$$

This is true for say $n = N^{3/2 - 2\gamma}$; we have something resembling 'good results' for a polynomial time in N, as opposed to $N \log N$ steps.

We could've done better here by using Exercise 10 and trying to optimise ν, but this already demonstrates that we can get the claimed polynomial dependence for this 'neutral' ODE.

The final regime, which for example the infection process is in, is 'stable', or even 'strongly stable'; trajectories tend to go towards a limit trajectory (if there is some relatively slow

convergence we'd say stable; if the convergence is exponentially fast, it's strongly stable). For simple one-dimensional, time-homogenous models like the infection process, the 'limit trajectory' will always be simply a constant function, say at Q. Here 'strongly stable' means something like $f(t, x) = Q - x$; again, this is 1-Lipschitz continuous but this time the trajectories are pushed 'as fast as possible' *towards* Q. For this kind of ODE, we can always use the 'catch-and-throw' trick of Exercise 12 to argue that the random process actually stays close to the ODE solution much longer (for exponentially long times); with some more work, one can write a general theorem that wraps the whole thing up in a single martingale analysis.

What's meant by 'behaviour on a domain $\mathcal{D}$' is that (usually!) a function $f(t, \mathbf{x})$ doesn't show this kind of simple global behaviour; if you draw the phase diagram of the ODE, it can be that in some regions trajectories come together fast and in others they move apart fast. However, if we look 'locally', we might see this behaviour: for example sufficiently near an attractive fixed point, the ODE will be (often strongly) stable, and we might take $\mathcal{D}$ to be such a neighbourhood.